

TACHS Practice Exam - Exam 4 - Part A
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Exam Part: A
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QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R4-S01
Section: Reading
Stimulus Type: Literary Passage
Title: The Missing Measure

Text:

For weeks, Jonah had practiced a short piano duet with his cousin Elise. At the community recital, they would play on the hall's old upright piano, whose keys felt heavier than those on Jonah's keyboard at home. He worried about the quick final section, so he practiced it until his hands could nearly move without thought.

Just before the recital, Elise opened the music and discovered that the last page was missing. Jonah searched their folders, the bench, and even beneath the piano. The audience settled into its seats while the program director quietly asked whether they needed to skip their turn.

Elise shook her head. She remembered the upper part, and Jonah knew most of the lower one. Still, he could not recall two measures near the end. Then he noticed a pattern: the uncertain measures echoed an earlier phrase, but began one step lower. He traced the pattern on the silent keys and nodded.

They began. The unfamiliar weight of the keys demanded attention, yet Jonah listened closely to Elise instead of racing ahead. When they reached the missing measures, the repeated pattern guided him. Their final chord was not perfectly simultaneous, but it was strong. Backstage, Jonah found the missing page tucked inside another piece. He laughed. The page had been useful during practice, but careful listening and understanding the music had carried them through the performance.

STIMULUS END

TACHS-R4-001
Section: Reading
Type: mcq
Category: Reading
Skill: Main Idea, Central Thought, or Summary
Stimulus ID: TACHS-R4-S01
Prompt:

Which statement best expresses the central idea of the passage?

- A) Understanding patterns and listening carefully help Jonah handle an unexpected problem.
- B) Musicians should always memorize every page before performing.

- C) An old piano prevents Jonah and Elise from playing well together.
- D) The program director wants the cousins to skip their recital.

TACHS-R4-002

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S01

Prompt:

Where was the missing page eventually found?

- A) Beneath the piano
- B) Inside another piece of music
- C) In the program director's folder
- D) Under a seat in the audience

TACHS-R4-003

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S01

Prompt:

What can reasonably be inferred about Jonah during the performance?

- A) He ignores Elise in order to concentrate on the keys.
- B) He decides that the final section is too difficult to attempt.
- C) He adjusts his approach instead of depending only on memorized movements.
- D) He discovers that the hall's piano is easier to play than his keyboard.

TACHS-R4-004

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R4-S01

Prompt:

How does the discovery of the repeated musical phrase affect the plot?

- A) It explains why the music page was placed in the wrong folder.
- B) It creates a disagreement between Jonah and Elise.
- C) It causes the program director to delay the recital.
- D) It provides Jonah with a way to reconstruct the measures he cannot remember.

STIMULUS START

Stimulus ID: TACHS-R4-S02

Section: Reading

Stimulus Type: Informational Passage

Title: When a Lake Turns Over

Text:

In many deep lakes, the water does not remain evenly mixed throughout the year. During summer, sunlight warms the upper layer. Because warm water is less dense than cold water, it stays near the surface. Deeper water remains cooler, and a narrow transition zone separates the two layers. Wind may stir the surface, but it usually cannot mix the entire lake.

As autumn air cools the surface, the upper water becomes denser. Eventually, its temperature approaches that of the deeper water. The density difference weakens, allowing wind to circulate water from top to bottom. This seasonal mixing is called turnover.

Turnover redistributes more than heat. During summer, tiny organisms near the surface may use dissolved nutrients, while decaying material carries nutrients toward the bottom. At the same time, deep water may receive little new oxygen from the air. When the lake mixes, oxygen moves downward and nutrients move upward.

This process can briefly make the water look cloudy because sediment and organic material are stirred. The cloudiness does not necessarily indicate pollution. In fact, turnover often supports the lake's living community by renewing access to oxygen and nutrients. A similar mixing period may occur in spring, after ice melts and before surface water becomes much warmer than deep water.

STIMULUS END

TACHS-R4-005

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S02

Prompt:

Why does warm surface water remain above colder water during summer?

- A) Warm water is less dense than cold water.
- B) Wind pushes all cold water to the lake bottom.
- C) Dissolved nutrients make surface water lighter.
- D) Sediment prevents the two layers from touching.

TACHS-R4-006

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S02

Prompt:

As used in the third paragraph, "redistributes" most nearly means

- A) measures for the first time
- B) spreads into new locations
- C) removes completely
- D) changes into another substance

TACHS-R4-007

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S02

Prompt:

Why does the author mention that turnover may make lake water appear cloudy?

- A) To argue that cloudy lakes should be closed to visitors
- B) To show that sediment is the main cause of lake pollution
- C) To clarify that a visible change can be part of a beneficial natural process
- D) To prove that autumn turnover is stronger than spring turnover

TACHS-R4-008

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S02

Prompt:

Which condition would most likely delay autumn turnover?

- A) Nutrients collecting near the lake bottom
- B) A brief increase in wind at the surface
- C) Organisms using oxygen in deep water
- D) Surface water remaining much warmer than deep water

STIMULUS START

Stimulus ID: TACHS-R4-S03

Section: Reading

Stimulus Type: Humanities Passage

Title: The Work Behind a Useful Map

Text:

A map may appear to be a simple picture of a place, but every map is also a collection of decisions. A cartographer must decide what to include, what to leave out, and how to represent features that cannot all fit at their actual size.

Those decisions depend on purpose. A subway map emphasizes station order and transfer points, even if the drawn routes bend differently from the tracks beneath the city. A hiking map gives more attention to elevation, water sources, and trail junctions. A map made for emergency planning may highlight evacuation routes and shelters while omitting details that would distract from quick decisions.

Scale creates another challenge. On a neighborhood map, individual streets and buildings may be useful. On a map of an entire country, those same features would crowd the page. Cartographers therefore generalize: they simplify coastlines, combine minor roads, or select only the most relevant landmarks.

These choices do not make a carefully designed map dishonest. Instead, they remind readers to ask what job the map was designed to perform. Trouble begins when a reader expects one

map to answer every possible question. The best map is not necessarily the one with the most information. It is the one whose selected information helps its intended users see the relationships they need.

STIMULUS END

TACHS-R4-009

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S03

Prompt:

Which statement best summarizes the passage?

- A) Effective maps select and simplify information according to a specific purpose.
- B) Subway maps are less accurate than maps used by hikers.
- C) Cartographers should include every landmark that can fit on a page.
- D) Map readers mainly need to understand how scale is calculated.

TACHS-R4-010

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S03

Prompt:

In the third paragraph, “generalize” most nearly means

- A) guess without evidence
- B) present in a simplified form
- C) enlarge every small feature
- D) arrange in alphabetical order

TACHS-R4-011

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S03

Prompt:

According to the passage, what does a subway map emphasize?

- A) The exact curves followed by underground tracks
- B) The elevation of each neighborhood
- C) Station order and transfer points
- D) Every street located above the subway

TACHS-R4-012

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S03

Prompt:

Which conclusion is best supported by the passage?

- A) A detailed map is useful for every type of traveler.
- B) Simplifying a map always makes it less trustworthy.
- C) Emergency maps should show the locations of all buildings.
- D) A map should be judged partly by how well it serves its intended users.

TACHS-R4-013

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S03

Prompt:

The author's primary purpose is to

- A) explain why mapmaking requires purposeful selection
- B) persuade readers to avoid using digital maps
- C) compare the careers of several cartographers
- D) describe how to measure distance with a map scale

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE4-S01

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Hallway Gallery

Text:

[1] Students at Monroe Middle School wanted more people to notice the artwork created in their classes. [2] They proposed turning an empty hallway into a rotating gallery. [3] First, the group measured the walls and asked the principal which hanging materials were permitted. [4] The cafeteria serves soup on Wednesdays. [5] Next, students designed reusable labels that listed each artist's name, medium, and brief statement. [6] Because the hallway leads to the auditorium, families can view the display before concerts and plays. [7] A new set of work will be installed every six weeks. [8] The gallery gives student artists a public audience while keeping the display fresh throughout the year.

STIMULUS END

TACHS-WE4-001

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S01

Prompt:

Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 6
- D) Sentence 7

TACHS-WE4-002

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S01

Prompt:

Which revision most effectively combines sentences 1 and 2?

- A) Students at Monroe Middle School wanted more people to notice their artwork, turning an empty hallway.
- B) Wanting artwork, an empty hallway was proposed by students as a rotating gallery.
- C) To help more people notice student artwork, students at Monroe Middle School proposed turning an empty hallway into a rotating gallery.
- D) Students wanted more people to notice artwork, they proposed an empty hallway gallery.

TACHS-WE4-003

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S01

Prompt:

Which word best replaces “brief” in sentence 5 without changing the intended meaning?

- A) concise
- B) tiny
- C) hurried
- D) narrow

TACHS-WE4-004

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S01

Prompt:

Which sentence would best follow sentence 7?

- A) Some students prefer painting, while others like sculpture.
- B) The school auditorium contains several rows of folding seats.
- C) Labels can be printed in more than one typeface.
- D) This schedule will allow many classes to share the limited wall space.

STIMULUS START

Stimulus ID: TACHS-WE4-S02

Section: Written Expression

Stimulus Type: Revision Passage

Title: Testing a Shade Sail

Text:

[1] The science club wondered whether a fabric shade sail could lower the temperature above the school's sunny courtyard tables. [2] Members placed identical thermometers above two tables at the same height. [3] One table was covered by the sail, while the other remained in direct sunlight. [4] The club recorded both temperatures every ten minutes for one hour. [5] Clouds covered the sun for several minutes during the test. [6] As a result, members repeated the experiment on two additional clear days. [7] The shaded table was cooler during most readings on all three clear-day trials. [8] The club presented its results to the principal and recommended a longer test in different seasons.

STIMULUS END

TACHS-WE4-005

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S02

Prompt:

Which sentence best states the purpose of the experiment and should remain as the opening?

- A) Sentence 2
- B) Sentence 1
- C) Sentence 4
- D) Sentence 7

TACHS-WE4-006

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S02

Prompt:

Which revision of sentence 3 uses correct and effective subordination?

- A) Although one table was covered by the sail, the other remained in direct sunlight.
- B) Covering one table with the sail and the other remained in direct sunlight.
- C) One table was covered by the sail, the other remained in direct sunlight.
- D) One table covered by the sail; while the other was in direct sunlight.

TACHS-WE4-007

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S02

Prompt:

Which revision of sentence 7 is most precise?

- A) The shaded table was kind of cooler in the readings.
- B) The table under the sail had temperatures and the other did too.
- C) The results were good for the shade sail on the tested days.
- D) On each clear-day trial, most readings at the shaded table were lower than those at the unshaded table.

TACHS-WE4-008

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S02

Prompt:

Which revision of sentence 8 correctly uses parallel structure?

- A) The club presented its results to the principal and recommended testing longer across different seasons.
- B) The club presented its results to the principal and recommending a longer test in different seasons.
- C) The club was presenting its results and a longer test was recommended.
- D) The club presented its results, and the recommendation of a longer seasonal test.

TACHS-WE4-009

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Neither of the two routes lead directly to the museum.
- B) Neither of the two routes leads directly to the museum.
- C) Neither of the two routes leading directly to the museum.
- D) Neither of the two routes have led direct to the museum.

TACHS-WE4-010

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses pronouns correctly?

- A) The coach gave the practice schedule to Elena and I.
- B) Me and Elena received the practice schedule from the coach.
- C) The coach gave Elena and me the practice schedule.
- D) The coach gave she and I the practice schedule.

TACHS-WE4-011

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: None

Prompt:

Which sentence is clearest and most concise?

- A) Due to the fact that the bridge was closed, we took a route that was different.
- B) The bridge being closed was the reason why our route was changed by us.
- C) We took another route, and this was because of the closed condition of the bridge.
- D) Because the bridge was closed, we took another route.

TACHS-WE4-012

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The collection of postcards was stored in a wooden box.

Line 2: Each card showed a landmark from a different city.

Line 3: The oldest cards were handled more careful than the others.

Line 4: Visitors could examine digital copies on a nearby screen.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE4-013

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: Before the workshop began, every participant received safety glasses.

Line 2: The instructor demonstrated how to secure the pieces firmly.

Line 3: Several students asked questions about the final assembly.

Line 4: After the demonstration, everyone return to a separate worktable.

- A) Line 1
- B) Line 4
- C) Line 2
- D) Line 3

SECTION 3 - MATHEMATICS

TACHS-M4-001

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A recipe uses $2\frac{1}{4}$ cups of oats for one batch. How many cups are needed for 3 batches?

A) $6\frac{3}{4}$

B) $6\frac{1}{4}$

C) $7\frac{1}{2}$

D) $8\frac{1}{4}$

TACHS-M4-002

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $18 - 3 \times (4 + 2)$?

A) -90

B) 0

C) 36

D) 90

TACHS-M4-003

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

At a fundraiser, 36 of 90 tickets were sold before noon. What percent of the tickets were sold?

A) 25%

B) 36%

C) 40%

D) 54%

TACHS-M4-004

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $5(x - 2) = 35$.

- A) 5
- B) 7
- C) 8
- D) 9

TACHS-M4-005

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The first four terms of a sequence are 4, 9, 14, 19. Which expression gives the n th term?

- A) $5n - 1$
- B) $4n + 1$
- C) $5n + 4$
- D) $n + 5$

TACHS-M4-006

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A streaming service charges a fixed fee of \$6 plus \$2 for each movie rented. Which equation gives the total cost c for renting m movies?

- A) $c = 6m + 2$
- B) $c = 2(m + 6)$
- C) $c = 2m + 6$
- D) $c = 8m$

TACHS-M4-007

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of books read by five students are 3, 5, 5, 7, and 10. What is the mean?

- A) 5
- B) 5.5
- C) 7
- D) 6

TACHS-M4-008

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 4 red, 3 blue, and 5 green tiles. One tile is chosen, replaced, and then another is chosen. What is the probability that both tiles are blue?

A) $\frac{1}{12}$

B) $\frac{1}{16}$

C) $\frac{1}{8}$

D) $\frac{1}{4}$

TACHS-M4-009

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Two angles are supplementary. One angle measures 118° . What is the measure of the other angle?

A) 62°

B) 72°

C) 118°

D) 242°

TACHS-M4-010

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A square has an area of 81 cm^2 . What is its perimeter?

A) 18 cm

B) 36 cm

C) 40 cm

D) 81 cm

TACHS-M4-011

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A cyclist travels 18 miles in 1.5 hours at a constant rate. What is the cyclist's average speed?

A) 9 miles per hour

B) 10 miles per hour

- C) 12 miles per hour
- D) 27 miles per hour

TACHS-M4-012

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A model uses a scale of 1 inch = 6 feet. A wall is 4.5 inches long on the model. How long is the actual wall?

- A) 10.5 feet
- B) 18 feet
- C) 24 feet
- D) 27 feet

SECTION 4 - ABILITY

TACHS-A4-001

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ○

Top-right: ○○○

Bottom-left: ■

Bottom-right: ?

Across each row, two copies of the same shape are added. Moving down changes an open shape to a filled square.

- A) ■■
- B) □□□
- C) ●●●
- D) ■■■

TACHS-A4-002

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ↑ | → | ↓

Row 2: → | ↓ | ←

Row 3: ↓ | ← | ?

Each step to the right or downward rotates the arrow 90° clockwise.

- A) ↑
- B) →
- C) ↓
- D) ←

TACHS-A4-003

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2 × 2 matrix.

Top-left: ○ inside □

Top-right: □ inside ○

Bottom-left: △ inside ◇

Bottom-right: ?

Across each row, the inner and outer shapes exchange positions.

- A) △ inside ◇
- B) ◇ inside △
- C) ▲ inside ◆
- D) ◇ beside △

TACHS-A4-004

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3 × 3 matrix.

Row 1: ○ | □ | ○□

Row 2: ● | ■ | ●■

Row 3: △ | ▲ | ?

In each row, the third cell places the first two symbols together in their original order.

- A) ▲△
- B) △■
- C) △▲
- D) ▲▲

TACHS-A4-005

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: open ○ above open □

Top-right: filled ● above filled ■

Bottom-left: open △ above open ◇

Bottom-right: ?

Across each row, positions stay fixed while both shapes change from open to filled.

A) filled ▲ above filled ◆

B) open △ above open ◇

C) filled ◆ above filled ▲

D) filled ▲ above open ◇

TACHS-A4-006

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular paper is folded in half from right to left. One hole is punched near the lower-left corner of the folded rectangle, away from all edges. After unfolding, where are the holes?

A) Near both lower corners, symmetric across the vertical centerline

B) Near both upper corners

C) One near the lower-left corner only

D) Four holes, one in each quadrant

TACHS-A4-007

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded from top to bottom and then from right to left. One hole is punched near the upper-right corner of the final folded square, away from all edges. What appears after unfolding?

A) Two holes near the top edge

B) Four holes, one in each quadrant, symmetric across both centerlines

C) Four holes along the left edge

D) Eight holes around the center

TACHS-A4-008

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded once along the diagonal from upper-right to lower-left. A small triangular notch is cut from the folded diagonal, away from its endpoints. What appears after unfolding?

- A) Two separate triangular notches on opposite outer edges
- B) One triangular notch at the upper-right corner
- C) A symmetric diamond-shaped opening centered on the diagonal
- D) Four triangular openings around the center

TACHS-A4-009

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded in half from left to right. The folded rectangle is then folded from bottom to top. A hole is punched near the center of the final folded square, away from all edges. After unfolding, how many holes appear?

- A) 1
- B) 2
- C) 8
- D) 4

TACHS-A4-010

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The arrangements ○|○, □|□, and △|△ share a rule. Which choice follows the same rule?

- A) ◇|◆
- B) ☆|☆
- C) ●|○
- D) +|×

TACHS-A4-011

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The figures ○●, □■, and △▲ share a rule. Which choice follows the same rule?

- A) ◆◇
- B) ☆○
- C) ◇◆
- D) ■□

TACHS-A4-012

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The arrangements “○ above □,” “△ above ◇,” and “☆ above +” share a rule. Which choice follows the same rule?

- A) ■ above ◆
- B) + above ☆
- C) ● beside □
- D) ◇ above ○

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R4-001 - Correct: A - Correct Answer: Understanding patterns and listening carefully help Jonah handle an unexpected problem. - Explanation: Jonah uses musical structure and attentive listening to perform successfully without the missing page.

TACHS-R4-002 - Correct: B - Correct Answer: Inside another piece of music - Explanation: Backstage, Jonah finds the missing page tucked inside another musical piece.

TACHS-R4-003 - Correct: C - Correct Answer: He adjusts his approach instead of depending only on memorized movements. - Explanation: Jonah attends to the unfamiliar keys and listens to Elise rather than relying solely on practiced motion.

TACHS-R4-004 - Correct: D - Correct Answer: It provides Jonah with a way to reconstruct the measures he cannot remember. - Explanation: Recognizing that the measures echo an earlier phrase lets Jonah determine how the missing portion should sound.

TACHS-R4-005 - Correct: A - Correct Answer: Warm water is less dense than cold water. - Explanation: The passage directly identifies the density difference as the reason warm water stays above cold water.

TACHS-R4-006 - Correct: B - Correct Answer: spreads into new locations - Explanation: Turnover moves oxygen downward and nutrients upward, spreading them to different parts of the lake.

TACHS-R4-007 - Correct: C - Correct Answer: To clarify that a visible change can be part of a beneficial natural process - Explanation: The author contrasts temporary cloudiness with turnover's benefits to prevent readers from assuming that cloudiness means pollution.

TACHS-R4-008 - Correct: D - Correct Answer: Surface water remaining much warmer than deep water - Explanation: A strong temperature and density difference would continue separating the layers and prevent full mixing.

TACHS-R4-009 - Correct: A - Correct Answer: Effective maps select and simplify information according to a specific purpose. - Explanation: The passage explains that mapmakers choose and generalize details to meet users' particular needs.

TACHS-R4-010 - Correct: B - Correct Answer: present in a simplified form - Explanation: The examples of simplified coastlines and combined minor roads define how cartographers generalize.

TACHS-R4-011 - Correct: C - Correct Answer: Station order and transfer points - Explanation: The second paragraph explicitly states that subway maps emphasize station sequence and transfers.

TACHS-R4-012 - Correct: D - Correct Answer: A map should be judged partly by how well it serves its intended users. - Explanation: The passage says the best map presents the relationships its intended users need.

TACHS-R4-013 - Correct: A - Correct Answer: explain why mapmaking requires purposeful selection - Explanation: The author describes how purpose and scale guide decisions about including, omitting, and simplifying information.

WRITTEN EXPRESSION ANSWERS

TACHS-WE4-001 - Correct: B - Correct Answer: Sentence 4 - Explanation: The cafeteria menu is unrelated to creating and maintaining the hallway art gallery.

TACHS-WE4-002 - Correct: C - Correct Answer: To help more people notice student artwork, students at Monroe Middle School proposed turning an empty hallway into a rotating gallery. - Explanation: This revision clearly connects the students' purpose to their proposal in one complete sentence.

TACHS-WE4-003 - Correct: A - Correct Answer: concise - Explanation: "Concise" precisely means brief while maintaining an appropriate formal tone.

TACHS-WE4-004 - Correct: D - Correct Answer: This schedule will allow many classes to share the limited wall space. - Explanation: The added sentence explains why rotating the work every six weeks benefits the school.

TACHS-WE4-005 - Correct: B - Correct Answer: Sentence 1 - Explanation: Sentence 1 introduces both the question being investigated and the shade sail being tested.

TACHS-WE4-006 - Correct: A - Correct Answer: Although one table was covered by the sail, the other remained in direct sunlight. - Explanation: "Although" correctly subordinates the first complete idea and connects it to the contrasting condition.

TACHS-WE4-007 - Correct: D - Correct Answer: On each clear-day trial, most readings at the shaded table were lower than those at the unshaded table. - Explanation: This revision states the comparison and testing conditions precisely without vague wording.

TACHS-WE4-008 - Correct: A - Correct Answer: The club presented its results to the principal and recommended testing longer across different seasons. - Explanation: The coordinated verbs "presented" and "recommended" are parallel and share the subject "club."

TACHS-WE4-009 - Correct: B - Correct Answer: Neither of the two routes leads directly to the museum. - Explanation: The singular subject "neither" requires the singular verb "leads."

TACHS-WE4-010 - Correct: C - Correct Answer: The coach gave Elena and me the practice schedule. - Explanation: "Elena and me" correctly uses the objective case for the people receiving the schedule.

TACHS-WE4-011 - Correct: D - Correct Answer: Because the bridge was closed, we took another route. - Explanation: This sentence expresses the cause and result directly without redundancy or unnecessary passive construction.

TACHS-WE4-012 - Correct: C - Correct Answer: Line 3 - Explanation: The verb "handled" must be modified by the adverb "carefully," not the adjective "careful."

TACHS-WE4-013 - Correct: B - Correct Answer: Line 4 - Explanation: The past-time context requires "returned," rather than the present-tense form "return."

MATHEMATICS ANSWERS

TACHS-M4-001 - Correct: A - Correct Answer: $6\frac{3}{4}$ - Explanation: $2\frac{1}{4} \times 3 = \frac{9}{4} \times 3 = \frac{27}{4} = 6\frac{3}{4}$.

TACHS-M4-002 - Correct: B - Correct Answer: 0 - Explanation: Evaluate parentheses and multiplication first: $4 + 2 = 6$, $3 \times 6 = 18$, and $18 - 18 = 0$.

TACHS-M4-003 - Correct: C - Correct Answer: 40% - Explanation: $36 \div 90 = 0.4$, and 0.4 equals 40%.

TACHS-M4-004 - Correct: D - Correct Answer: 9 - Explanation: Divide by 5 to get $x - 2 = 7$, then add 2 to obtain $x = 9$.

TACHS-M4-005 - Correct: A - Correct Answer: $5n - 1$ - Explanation: The sequence increases by 5, and $5(1) - 1 = 4$, the first term.

TACHS-M4-006 - Correct: C - Correct Answer: $c = 2m + 6$ - Explanation: The variable charge is \$2 times m movies, and the fixed \$6 fee is then added.

TACHS-M4-007 - Correct: D - Correct Answer: 6 - Explanation: The values total 30, and $30 \div 5 = 6$.

TACHS-M4-008 - Correct: B - Correct Answer: $\frac{1}{16}$ - Explanation: The chance of blue on each draw is $\frac{3}{12} = \frac{1}{4}$, so $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.

TACHS-M4-009 - Correct: A - Correct Answer: 62° - Explanation: Supplementary angles total 180° , so $180^\circ - 118^\circ = 62^\circ$.

TACHS-M4-010 - Correct: B - Correct Answer: 36 cm - Explanation: A square with area 81 cm^2 has side length 9 cm, so its perimeter is $4 \times 9 = 36\text{ cm}$.

TACHS-M4-011 - Correct: C - Correct Answer: 12 miles per hour - Explanation: Average speed equals distance divided by time: $18 \div 1.5 = 12$ miles per hour.

TACHS-M4-012 - Correct: D - Correct Answer: 27 feet - Explanation: Multiply the model length by the scale: $4.5 \times 6 = 27$ feet.

ABILITY ANSWERS

TACHS-A4-001 - Correct: D - Correct Answer: ■■■ - Explanation: Moving right adds two copies, and moving down changes the open circle family to filled squares.

TACHS-A4-002 - Correct: A - Correct Answer: ↑ - Explanation: Rotating ← 90° clockwise produces ↑, continuing the pattern across both the row and column.

TACHS-A4-003 - Correct: B - Correct Answer: ◇ inside Δ - Explanation: The inner and outer shapes exchange positions across each row without changing their forms.

TACHS-A4-004 - Correct: C - Correct Answer: Δ▲ - Explanation: Each third cell combines the first and second symbols in the same left-to-right order.

TACHS-A4-005 - Correct: A - Correct Answer: filled ▲ above filled ◆ - Explanation: Both bottom-row shapes keep their positions and change from open to their corresponding filled forms.

TACHS-A4-006 - Correct: A - Correct Answer: Near both lower corners, symmetric across the vertical centerline - Explanation: Undoing the horizontal half-fold reflects the lower-left punch to the lower-right side.

TACHS-A4-007 - Correct: B - Correct Answer: Four holes, one in each quadrant, symmetric across both centerlines - Explanation: Reversing two perpendicular folds reflects the punch across both centerlines, producing four symmetric locations.

TACHS-A4-008 - Correct: C - Correct Answer: A symmetric diamond-shaped opening centered on the diagonal - Explanation: The triangular notch on the fold reflects across the diagonal to form a symmetric diamond-shaped opening.

TACHS-A4-009 - Correct: D - Correct Answer: 4 - Explanation: Reversing two perpendicular folds reflects the interior punch across both centerlines, producing four holes.

TACHS-A4-010 - Correct: B - Correct Answer: ☆|☆ - Explanation: Each example places two identical open symbols on opposite sides of a vertical line.

TACHS-A4-011 - Correct: C - Correct Answer: ◇◆ - Explanation: Each pair contains an open shape followed by the filled form of that same shape.

TACHS-A4-012 - Correct: D - Correct Answer: ◇ above ○ - Explanation: Like each example, the choice places two different open symbols in a vertical arrangement.

TACHS Practice Exam - Exam 4 - Part B
Filename: tachs_full_exam_04_part_b.txt
Exam Family: TACHS
Exam Number: 4
Exam Part: B
Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R4-S04

Section: Reading

Stimulus Type: Literary or Narrative Passage

Title: The Reversed Poster

Text:

Leah volunteered to print a large welcome poster for the school language fair. The art room had a small hand press, and she arranged wooden letters in a frame exactly as the words should appear. When she lifted the first sheet, the message was backward.

Mr. Okafor did not reset the letters for her. Instead, he placed a mirror beside the frame. In the reflection, Leah could read the arrangement correctly. She realized that the raised letters touched the paper face down, so the frame had to hold a reversed version of the final message.

Leah began again, checking each line in the mirror before tightening the frame. The second proof was readable, but the spaces between words were uneven. She inserted thin blank blocks, printed another proof, and compared the margins on both sides. Each test used more paper, yet every proof revealed a different problem while it was still easy to fix.

On the morning of the fair, the finished poster hung above the entrance. Visitors saw a clean message in three languages. Leah noticed something else: the poster looked simple only because the complicated decisions had been settled before the final sheet was printed.

STIMULUS END

TACHS-R4-014

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S04

Prompt:

What tool helps Leah check whether the letters are arranged correctly?

- A) A ruler beside the press
- B) A mirror beside the frame
- C) A lamp beneath the paper
- D) A photograph of the entrance

TACHS-R4-015

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S04

Prompt:

Why does Leah print several proofs before making the final poster?

- A) She wants to display every version at the fair.
- B) She must use all the paper in the art room.
- C) She is unsure which languages belong on the poster.
- D) Each proof reveals a problem that can still be corrected.

TACHS-R4-016

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S04

Prompt:

Which statement best expresses the main idea of the passage?

- A) A polished result often depends on careful testing and correction.
- B) Hand presses are faster than modern printers for large posters.
- C) School fairs should display information in several languages.
- D) Mirrors are the most important tools in an art room.

TACHS-R4-017

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S04

Prompt:

As used in the third paragraph, "proof" most nearly means

- A) evidence that an argument is true.
- B) a protective coating placed on paper.
- C) a trial copy made for checking.
- D) a rule that cannot be changed.

STIMULUS START

Stimulus ID: TACHS-R4-S05

Section: Reading

Stimulus Type: Scientific or Technical Passage

Title: Why Owl Wings Sound Quiet

Text:

Many birds create a noticeable rush of air as their wings move. Owls, however, can approach prey with unusually little flight noise. Their silence does not come from one feature alone. Several parts of an owl feather work together to change how air passes over the wing.

Along the front edge of a primary wing feather, small comb-like projections divide a strong stream of air into many smaller currents. A soft, velvety surface on the feather absorbs some of the tiny vibrations created by those currents. The trailing edge is fringed rather than sharply cut, which allows air to leave the wing more gradually.

These features reduce noise, but they also create trade-offs. The soft feather surface is not ideal for high-speed flight in heavy rain, and most owls do not travel as rapidly as some daytime hunters. Quiet movement is especially useful during short flights when an owl must listen for faint sounds below.

Engineers study owl feathers when designing quieter fans, turbines, and aircraft parts. A useful design does not simply copy the shape of a feather. It identifies how separated airflow, soft surfaces, and gradual edges reduce sound, then adapts those principles to a different material and purpose.

STIMULUS END

TACHS-R4-018

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S05

Prompt:

What do the comb-like projections on an owl feather do?

- A) Divide a strong air stream into smaller currents
- B) Hold rainwater away from the wing
- C) Increase the bird's speed during long flights
- D) Make the feather surface harder

TACHS-R4-019

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S05

Prompt:

Why does the author mention fans, turbines, and aircraft parts?

- A) To show that owl feathers are made from industrial materials
- B) To argue that machines should move as slowly as owls
- C) To explain how biological principles can inspire quieter technology
- D) To prove that all flight noise can be eliminated

TACHS-R4-020

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R4-S05

Prompt:

How is the passage mainly organized?

- A) It compares several owl species and ranks their hunting success.
- B) It gives instructions for building a model wing.
- C) It presents a mystery and then rejects every possible explanation.
- D) It explains several feather features, their effects, and possible applications.

TACHS-R4-021

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S05

Prompt:

As used in the third paragraph, "trade-offs" most nearly means

- A) exchanges of feathers between birds.
- B) advantages gained with accompanying disadvantages.
- C) changes that occur only during rain.
- D) measurements used to compare flight speed.

STIMULUS START

Stimulus ID: TACHS-R4-S06

Section: Reading

Stimulus Type: Informational or Historical-Style Passage

Title: When Elevators Learned to Show Their Location

Text:

Early passenger elevators moved people between floors, but riders often had little information about where the car was. An operator controlled the motion and announced each stop. In a busy building, a person waiting in the hallway might not know whether the elevator was one floor away or near the roof.

As buildings grew taller, manufacturers added position indicators. Some early systems used a moving pointer beside a vertical list of floor numbers. Others placed small lamps above the doors. The elevator car activated a switch as it passed each floor, causing the matching lamp to glow.

The indicator did more than satisfy curiosity. People waiting could decide whether to remain at one elevator or walk to another. Maintenance workers could also notice when the displayed position did not match the car's actual location. Later systems added direction arrows, arrival signals, and spoken announcements.

Modern digital displays look different from the mechanical pointers of earlier buildings, but they solve the same communication problem. The machine is moving inside a closed shaft, while the people using it need clear information outside that shaft. A small display connects those two situations by making hidden motion understandable.

STIMULUS END

TACHS-R4-022

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S06

Prompt:

How did some early elevator indicators show the car's position?

- A) By opening a window into the elevator shaft
- B) By printing a new ticket at each floor
- C) By moving a pointer or lighting a floor lamp
- D) By slowing the car near every hallway

TACHS-R4-023

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S06

Prompt:

Why did position indicators become more useful as buildings grew taller?

- A) The elevator could be farther away and harder to track without information.
- B) Tall buildings required every rider to operate the elevator.
- C) Mechanical pointers worked only above a certain height.
- D) People in tall buildings were not allowed to use stairs.

TACHS-R4-024

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S06

Prompt:

What is the author's main purpose in the final paragraph?

- A) To criticize digital displays for replacing mechanical devices
- B) To show that old and new indicators address the same need for information
- C) To explain how to repair an elevator position switch
- D) To argue that elevator shafts should be visible

TACHS-R4-025

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S06

Prompt:

Which statement best summarizes the passage?

- A) Elevator operators disappeared as soon as electric lamps were invented.
- B) Tall buildings were designed mainly to test new communication devices.
- C) Mechanical pointers are more accurate than modern digital displays.
- D) Position indicators made hidden elevator movement clearer and more useful to people.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE4-S03

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Reusable Container Return Station

Text:

[1] Our student council tested a return station for reusable lunch containers. [2] Students placed empty containers on a sloped rack beside the cafeteria exit. [3] The rack guided each container toward one of three labeled bins. [4] During the first lunch period, several students paused because the labels were printed too small. [5] We replaced them with larger labels and added a sample container above each bin. [6] After the change, the line moved more quickly. [7] The

station collected 214 containers in one week. [8] Clear directions made the return process easier for students and cafeteria workers.

STIMULUS END

TACHS-WE4-014

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S03

Prompt:

Which sentence would best follow sentence 3 to explain how the bins were used?

- A) The cafeteria serves lunch during three periods.
- B) Some containers were round, while others were square.
- C) One bin held cups, one held bowls, and one held divided trays.
- D) The council meets twice each month after school.

TACHS-WE4-015

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S03

Prompt:

Which revision of sentence 4 is most effective?

- A) During the first lunch period, several students paused because the labels were too small to read quickly.
- B) During the first lunch period several students paused, the labels were printed too small.
- C) Several students, during the first lunch period, because the labels were printed too small.
- D) The labels printed too small, and several students during the first lunch period paused.

TACHS-WE4-016

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S03

Prompt:

Which revision of sentence 6 is most precise?

- A) After that, things improved a lot.
- B) Then the whole situation became better for everyone.
- C) The change was nice, and the line did something different.
- D) After the larger labels were added, students sorted containers with fewer delays.

TACHS-WE4-017

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S03

Prompt:

Which sentence best concludes the passage if sentence 8 is replaced?

- A) Reusable containers are sold in many sizes and colors.
- B) The trial showed that a return system works best when its categories are immediately clear.
- C) Some students brought lunches from home during the week.
- D) The cafeteria exit is also near the main staircase.

STIMULUS START

Stimulus ID: TACHS-WE4-S04

Section: Written Expression

Stimulus Type: Revision Passage

Title: Training the Morning Announcers

Text:

[1] Four students volunteered to deliver the school's morning announcements. [2] Each student

practiced reading the same short script into a microphone. [3] The recordings were reviewed for pace, volume, and clear pronunciation. [4] One speaker rushed through dates, while another paused so often that the message sounded disconnected. [5] After listening, the students marked places to breathe and words that needed emphasis. [6] They recorded the script again, and every version was easier to understand. [7] The microphone stand was made of black metal. [8] The exercise showed that effective speaking requires both preparation and careful listening.
STIMULUS END

TACHS-WE4-018

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S04

Prompt:

'Where should the following sentence be placed?

"The group also noted whether each speaker sounded confident rather than memorized."

- A) Before sentence 1
- B) After sentence 1
- C) After sentence 3
- D) After sentence 8

TACHS-WE4-019

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S04

Prompt:

Which revision most effectively combines sentences 4 and 5?

- A) One speaker rushed through dates, another paused often, the students marked the script.
- B) After hearing that one speaker rushed and another paused too often, the students marked breathing points and words for emphasis.
- C) The students marked the script, one speaker rushed, and another paused.
- D) Rushing through dates and pausing often, the script was marked by the students.

TACHS-WE4-020

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S04

Prompt:

Which sentence should be deleted because it does not support the passage's focus?

- A) Sentence 7
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

TACHS-WE4-021

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE4-S04

Prompt:

Which revision of sentence 2 uses standard written US English?

- A) Each student practice reading the same short script.
- B) Each student practiced reading the same short script into a microphone.
- C) Each student have practiced reading the same short script.
- D) Each student were practicing the same short script.

TACHS-WE4-022

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses the pronoun correctly?

- A) The advisor spoke with Maya and I after the meeting.
- B) The advisor spoke with Maya and myself after the meeting.
- C) The advisor spoke with Maya and me after the meeting.
- D) The advisor spoke with she and Maya after the meeting.

TACHS-WE4-023

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Neither of the two signs is visible from the doorway.
- B) Neither of the two signs are visible from the doorway.
- C) Neither of the two sign is visible from the doorway.
- D) Neither of the two signs were visible from the doorway now.

TACHS-WE4-024

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The committee reviewed the three proposals.

Line 2: Each proposal included a cost estimate.

Line 3: The members discussed the options calmly.

Line 4: One of the plans require additional storage.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE4-025

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The boxes arrived on Tuesday.

Line 2: Volunteers opened them carefully.

Line 3: The labels were bright and clearly.

Line 4: Every item was counted before storage.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE4-026

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: Before the doors opened, the guides checked their notes.

Line 2: Every guide carried a printed schedule.

Line 3: The visitors asked several useful questions.

Line 4: Later, the guides answers were collected.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M4-013

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $|-18| - |7 - 12|$?

A) 5

B) 11

C) 12

D) 13

TACHS-M4-014

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $5/6 - 1/4 + 1/3$?

A) $7/12$

B) $11/12$

C) $13/12$

D) $17/12$

TACHS-M4-015

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

How many positive factors does 36 have?

A) 9

B) 8

C) 10

D) 12

TACHS-M4-016

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A line passes through the points (2, 5) and (6, 17). What is its rate of change?

A) 2

B) 2.5

C) 3

D) 4

TACHS-M4-017

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Which expression is equivalent to $3(2x - 5) + 4x$?

- A) $6x - 1$
- B) $7x - 15$
- C) $10x + 15$
- D) $10x - 15$

TACHS-M4-018

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A frequency list contains one 2, three 3s, five 4s, and one 5. What is the mean?

- A) 3.6
- B) 3.8
- C) 4
- D) 4.2

TACHS-M4-019

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

One letter is selected from the word MATH, and one number is selected from 1, 2, and 3. What is the probability of selecting a vowel and an even number?

- A) $1/6$
- B) $1/12$
- C) $1/4$
- D) $1/3$

TACHS-M4-020

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The ordered pairs (1, 12), (2, 18), (3, 25), and (4, 31) compare hours practiced with points earned. Which conclusion is best supported?

- A) Points decrease as practice time increases.
- B) Practice time and points have no relationship.
- C) Every additional hour always adds exactly 6 points.
- D) The data show a positive association between practice time and points.

TACHS-M4-021

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Three vertices of a rectangle are (1, 2), (1, 7), and (6, 2). Which point is the fourth vertex?

- A) (2, 6)

- B) (5, 7)
- C) (6, 7)
- D) (7, 6)

TACHS-M4-022

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A 90° central angle cuts a circle into a sector. What fraction of the circle is the sector?

- A) $\frac{1}{2}$
- B) $\frac{1}{4}$
- C) $\frac{1}{3}$
- D) $\frac{3}{4}$

TACHS-M4-023

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A cylinder has a radius of 3 centimeters and a height of 5 centimeters. Using 3.14 for π , what is its volume?

- A) 141.3 cubic centimeters
- B) 94.2 cubic centimeters
- C) 47.1 cubic centimeters
- D) 282.6 cubic centimeters

TACHS-M4-024

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A faucet fills 9 liters in 6 minutes at a constant rate. How many milliliters does it fill in 4 minutes?

- A) 4,500 milliliters
- B) 5,400 milliliters
- C) 6,000 milliliters
- D) 7,500 milliliters

TACHS-M4-025

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

Use 1 square meter = 10,000 square centimeters. How many square centimeters are equal to 2.5 square meters?

- A) 250
- B) 2,500
- C) 20,000
- D) 25,000

SECTION 4 - ABILITY

TACHS-A4-013

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: $\uparrow \circ$

Top-right: $\rightarrow \bullet$

Bottom-left: $\leftarrow \square$

Bottom-right: ?

Moving right rotates the arrow 90° clockwise and changes the open shape to filled.

A) $\uparrow \blacksquare$

B) $\downarrow \square$

C) $\downarrow \blacksquare$

D) $\leftarrow \blacksquare$

TACHS-A4-014

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×3 matrix.

Row 1: $\infty \mid \infty \mid \bullet$

Row 2: $\square \mid \square \mid ?$

In each row, subtract the first count from the second count, then fill the remaining shapes.

A) $\square$

B) $\blacksquare$

C) $\blacksquare$

D) $\blacksquare$

TACHS-A4-015

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: $\circ \mid \square \mid \triangle\triangle$

Row 2: $\square \mid \triangle\triangle \mid \circ$

Row 3: $\triangle\triangle \mid \circ \mid ?$

Each row shifts the three cells one position left.

A) ∞

B) $\triangle\triangle$

C) $\square$

D) $\square$

TACHS-A4-016

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: Top-left $\circ$

Top-right: Top-right $\bullet$

Bottom-left: Bottom-left $\square$

Bottom-right: ?

Moving right shifts the symbol horizontally and fills it.

A) Top-left $\blacksquare$

B) Bottom-right $\blacksquare$

C) Bottom-right $\square$

D) Top-right ■

TACHS-A4-017

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from bottom to top and then from right to left. One hole is punched near the lower-right corner of the final folded square, away from all edges and folds. What appears after unfolding?

- A) Two holes in a vertical pair
- B) Two holes in a horizontal pair
- C) Three holes forming an L
- D) Four holes in a 2 × 2 arrangement

TACHS-A4-018

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is divided into four equal vertical panels and folded accordion-style into one panel. One interior hole is punched through the stack. What appears after unfolding?

- A) Four holes in one horizontal row
- B) Four holes in one vertical column
- C) Eight holes in two rows
- D) Two holes near the center

TACHS-A4-019

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded once along the diagonal from the upper-left corner to the lower-right corner. One hole is punched near the folded triangle's upper-left side, away from the fold. What appears after unfolding?

- A) One hole on the diagonal
- B) Two holes reflected across the diagonal
- C) Four holes near the corners
- D) Two holes reflected across a vertical line

TACHS-A4-020

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from left to right. A semicircular notch is cut at the midpoint of the folded edge created by the fold. What appears after unfolding?

- A) Two notches on the outer edges
- B) Four holes around the center
- C) One circular hole centered on the vertical midline
- D) One semicircular notch on the left edge

TACHS-A4-021

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

● ○ ●

■ □ ■

▲ △ ▲

Which choice follows the same rule?

A) ◆ ◇ ◆

B) ◇ ◆ ◇

C) ◆ ◆ ◇

D) ◇ ◇ ◆

TACHS-A4-022

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share the same rule:

○ | ○

□ - □

△ / △

Which choice follows the same rule?

A) ◇ ◆ ◇

B) ◇ ◇ |

C) ◆ \ ◆

D) ◇ \ ◇

TACHS-A4-023

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

● ○ ○

■ □ □

▲ △ △

Which choice follows the same rule?

A) ◇ ◆ ◆

B) ◆ ◇ ◆

C) ◆ ◇ ◇

D) ◇ ◇ ◆

TACHS-A4-024

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share the same rule:

↑ → ↓

→ ↓ ←

↓ ← ↑

Which choice follows the same rule?

A) ↑ ↓ ←

B) ← ↑ →

C) ← → ↑

D) → ↑ ←

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R4-014 - Correct: B - Correct Answer: A mirror beside the frame - Explanation: The mirror shows how the reversed letter arrangement will appear when printed.

TACHS-R4-015 - Correct: D - Correct Answer: Each proof reveals a problem that can still be corrected. - Explanation: The proofs expose reversed text, uneven spacing, and margin problems before the final printing.

TACHS-R4-016 - Correct: A - Correct Answer: A polished result often depends on careful testing and correction. - Explanation: Leah produces a simple-looking final poster by checking and correcting several hidden details.

TACHS-R4-017 - Correct: C - Correct Answer: a trial copy made for checking. - Explanation: In printing, a proof is a test copy examined before the final version is produced.

TACHS-R4-018 - Correct: A - Correct Answer: Divide a strong air stream into smaller currents - Explanation: The passage states that the projections split a strong stream of air into smaller currents.

TACHS-R4-019 - Correct: C - Correct Answer: To explain how biological principles can inspire quieter technology - Explanation: The examples show how engineers adapt noise-reducing principles from owl feathers to machines.

TACHS-R4-020 - Correct: D - Correct Answer: It explains several feather features, their effects, and possible applications. - Explanation: The passage moves from feather structures to their effects, trade-offs, and engineering uses.

TACHS-R4-021 - Correct: B - Correct Answer: advantages gained with accompanying disadvantages. - Explanation: The quiet-flight features provide benefits but also reduce performance in some conditions.

TACHS-R4-022 - Correct: C - Correct Answer: By moving a pointer or lighting a floor lamp - Explanation: The passage describes moving pointers and lamps activated as the car passed each floor.

TACHS-R4-023 - Correct: A - Correct Answer: The elevator could be farther away and harder to track without information. - Explanation: More floors increased the possible distance and waiting uncertainty, making location information more valuable.

TACHS-R4-024 - Correct: B - Correct Answer: To show that old and new indicators address the same need for information - Explanation: The final paragraph connects modern and early displays by emphasizing their shared communication purpose.

TACHS-R4-025 - Correct: D - Correct Answer: Position indicators made hidden elevator movement clearer and more useful to people. - Explanation: The passage traces how indicators communicate elevator location to waiting riders and workers.

WRITTEN EXPRESSION ANSWERS

TACHS-WE4-014 - Correct: C - Correct Answer: One bin held cups, one held bowls, and one held divided trays. - Explanation: The added sentence directly explains the three labeled container categories introduced in sentence 3.

TACHS-WE4-015 - Correct: A - Correct Answer: During the first lunch period, several students paused because the labels were too small to read quickly. - Explanation: Choice A clearly connects the students' pause with the labels' limited readability in a complete sentence.

TACHS-WE4-016 - Correct: D - Correct Answer: After the larger labels were added, students sorted containers with fewer delays. - Explanation: Choice D identifies the change and states its observable effect precisely.

TACHS-WE4-017 - Correct: B - Correct Answer: The trial showed that a return system works best when its categories are immediately clear. - Explanation: Choice B connects the test results to the passage's central lesson about clear directions.

TACHS-WE4-018 - Correct: C - Correct Answer: After sentence 3 - Explanation: The sentence adds another review criterion and therefore belongs after the list in sentence 3.

TACHS-WE4-019 - Correct: B - Correct Answer: After hearing that one speaker rushed and another paused too often, the students marked breathing points and words for emphasis. - Explanation: Choice B uses a clear introductory clause and correctly identifies the students as the people who revised the script.

TACHS-WE4-020 - Correct: A - Correct Answer: Sentence 7 - Explanation: The material of the microphone stand does not contribute to the discussion of speaking practice.

TACHS-WE4-021 - Correct: B - Correct Answer: Each student practiced reading the same short script into a microphone. - Explanation: The singular subject "each student" correctly takes the singular past-tense verb "practiced."

TACHS-WE4-022 - Correct: C - Correct Answer: The advisor spoke with Maya and me after the meeting. - Explanation: The preposition "with" requires the objective pronoun "me."

TACHS-WE4-023 - Correct: A - Correct Answer: Neither of the two signs is visible from the doorway.
 - Explanation: The singular subject "neither" requires the singular verb "is."
 TACHS-WE4-024 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 contains the error: the singular subject "one" requires "requires."
 TACHS-WE4-025 - Correct: C - Correct Answer: Line 3 - Explanation: Line 3 contains the error: the adjective "clear," not the adverb "clearly," should follow "were."
 TACHS-WE4-026 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 contains the error: the plural possessive form should be "guides' answers."

MATHEMATICS ANSWERS

TACHS-M4-013 - Correct: D - Correct Answer: 13 - Explanation: The absolute values are 18 and 5, so $18 - 5 = 13$.
 TACHS-M4-014 - Correct: B - Correct Answer: $11/12$ - Explanation: Using twelfths gives $10/12 - 3/12 + 4/12 = 11/12$.
 TACHS-M4-015 - Correct: A - Correct Answer: 9 - Explanation: Since $36 = 2^2 \times 3^2$, it has $(2 + 1)(2 + 1) = 9$ positive factors.
 TACHS-M4-016 - Correct: C - Correct Answer: 3 - Explanation: The rate of change is $(17 - 5) \div (6 - 2) = 12 \div 4 = 3$.
 TACHS-M4-017 - Correct: D - Correct Answer: $10x - 15$ - Explanation: Distribute and combine like terms: $6x - 15 + 4x = 10x - 15$.
 TACHS-M4-018 - Correct: A - Correct Answer: 3.6 - Explanation: The total is $2 + 9 + 20 + 5 = 36$ across 10 values, so the mean is 3.6.
 TACHS-M4-019 - Correct: B - Correct Answer: $1/12$ - Explanation: Only A is a vowel and only 2 is even, so the probability is $1/4 \times 1/3 = 1/12$.
 TACHS-M4-020 - Correct: D - Correct Answer: The data show a positive association between practice time and points. - Explanation: The point values generally rise as practice hours increase, showing a positive association.
 TACHS-M4-021 - Correct: C - Correct Answer: (6, 7) - Explanation: The missing vertex must share $x = 6$ with one point and $y = 7$ with another, giving (6, 7).
 TACHS-M4-022 - Correct: B - Correct Answer: $1/4$ - Explanation: The sector represents $90/360$ of the circle, which simplifies to $1/4$.
 TACHS-M4-023 - Correct: A - Correct Answer: 141.3 cubic centimeters - Explanation: Volume is $\pi r^2 h = 3.14 \times 3^2 \times 5 = 141.3$ cubic centimeters.
 TACHS-M4-024 - Correct: C - Correct Answer: 6,000 milliliters - Explanation: The rate is 1.5 liters per minute, so 4 minutes gives 6 liters, or 6,000 milliliters.
 TACHS-M4-025 - Correct: D - Correct Answer: 25,000 - Explanation: Multiply 2.5 by 10,000 to get 25,000 square centimeters.

ABILITY ANSWERS

TACHS-A4-013 - Correct: A - Correct Answer: $\uparrow \blacksquare$ - Explanation: Rotating $\leftarrow$ clockwise gives $\uparrow$, and changing $\square$ to its filled form gives $\blacksquare$.
 TACHS-A4-014 - Correct: C - Correct Answer: $\blacksquare \blacksquare$ - Explanation: Five squares minus three squares leaves two, and the result is filled: $\blacksquare \blacksquare$.
 TACHS-A4-015 - Correct: D - Correct Answer: $\square \square$ - Explanation: The third row continues the shift with $\triangle \triangle$, $\circ$, then $\square \square$.
 TACHS-A4-016 - Correct: B - Correct Answer: Bottom-right $\blacksquare$ - Explanation: The bottom-left open square shifts to bottom-right and changes to a filled square.
 TACHS-A4-017 - Correct: D - Correct Answer: Four holes in a 2×2 arrangement - Explanation: Two perpendicular half-folds create four layers, so the punch reflects into four quadrant positions.
 TACHS-A4-018 - Correct: A - Correct Answer: Four holes in one horizontal row - Explanation: Each of the four stacked panels receives the punch, producing one hole in each panel across the sheet.
 TACHS-A4-019 - Correct: B - Correct Answer: Two holes reflected across the diagonal - Explanation: A single diagonal fold creates one reflected partner across that diagonal.
 TACHS-A4-020 - Correct: C - Correct Answer: One circular hole centered on the vertical midline - Explanation: The semicircle lies against the fold, so its reflected half forms one complete circular hole on the midline.
 TACHS-A4-021 - Correct: A - Correct Answer: $\blacklozenge \diamond \blacklozenge$ - Explanation: Each group places one open shape between two filled copies of that same shape.
 TACHS-A4-022 - Correct: D - Correct Answer: $\diamond \setminus \diamond$ - Explanation: Each valid group places one line between two identical open shapes.
 TACHS-A4-023 - Correct: C - Correct Answer: $\blacklozenge \diamond \blacklozenge$ - Explanation: Each arrangement contains one filled shape followed by two open copies of that same shape.
 TACHS-A4-024 - Correct: B - Correct Answer: $\leftarrow \uparrow \rightarrow$ - Explanation: Each group lists three consecutive directions in clockwise order.

TACHS Practice Exam - Exam 4 - Part B

Filename: tachs_full_exam_04_part_b.txt

Exam Family: TACHS

Exam Number: 4

Exam Part: B

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R4-S04

Section: Reading

Stimulus Type: Literary Passage

Title: The Uneven Glaze

Text:

Mara opened the pottery kiln expecting twelve glossy blue bowls. Eleven looked as planned, but the last was streaked with pale lines. She had measured the glaze carefully, so she assumed she had made a mistake while applying it.

Mr. Chen examined the bowl without declaring it ruined. He asked Mara where it had stood in the kiln. She pointed to a shelf near the door. Then they compared the bowl with two test tiles fired in the same area. Both tiles showed faint pale marks.

“Perhaps the glaze is telling us something about the kiln,” Mr. Chen said. He explained that a loose seal could allow cool air to enter near the door. The changing temperature might keep the glaze from melting evenly.

Mara wanted to hide the bowl before the student art sale. Instead, she placed it beside the smooth blue ones with a card describing what she had learned. Visitors paused at the unusual bowl more often than at any other. One customer said the streaks looked like winter branches and bought it.

The sale pleased Mara, but the compliment was not the day’s most useful result. On Monday, she helped Mr. Chen test the kiln door and replace its worn seal. The next firing heated evenly. Mara kept a photograph of the streaked bowl above her worktable, a reminder that an imperfect result can become evidence when someone studies it closely.

STIMULUS END

TACHS-R4-014

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S04

Prompt:

What do the two test tiles reveal?

- A) Pale marks also appeared on objects fired near the kiln door.
- B) Mara used a different glaze on the final bowl.
- C) The kiln became hotter near its center.
- D) Mr. Chen had tested the wrong shelf.

TACHS-R4-015

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S04

Prompt:

Why does Mr. Chen ask where the bowl stood in the kiln?

- A) He wants to choose a better place to display it.
- B) He suspects its position may help explain the streaks.
- C) He needs to count the objects on each shelf.
- D) He thinks Mara forgot which glaze she applied.

TACHS-R4-016

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S04

Prompt:

Which statement best expresses a central idea of the passage?

- A) Handmade objects should always look exactly alike.
- B) Customers value unusual objects more than useful ones.
- C) Examining an unexpected result can turn a flaw into useful information.
- D) Artists should sell their work before checking it carefully.

TACHS-R4-017

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S04

Prompt:

As used in the final paragraph, “evidence” most nearly means

- A) a decoration added to improve an object.
- B) a promise that a result will occur again.
- C) a reward for completing difficult work.
- D) information that helps support an explanation.

STIMULUS START

Stimulus ID: TACHS-R4-S05

Section: Reading

Stimulus Type: Scientific Passage

Title: How Fog Collectors Gather Water

Text:

In some dry coastal regions, rain is scarce even though thick fog frequently moves across hillsides. Fog collectors can capture some of the water carried in that air. A typical collector consists of a fine mesh panel held upright by a frame. As wind pushes fog through the mesh, tiny droplets strike the threads and join together. Once the drops become heavy enough, gravity pulls them into a channel below the panel. From there, the water flows into a storage container.

The amount collected depends on several factors. A site needs regular fog and enough wind to move moist air through the mesh. The threads must be spaced closely enough to catch droplets but not so closely that they block airflow. Researchers also test different materials and coatings to help drops slide downward rather than remain on the panel.

Fog collectors do not create water, and they cannot supply every community with all the water it needs. Their usefulness varies by location and season. Still, they can provide water for gardens, tree seedlings, or other limited purposes where suitable conditions exist. Because the panels require no fuel to operate, they may serve as a simple addition to other water sources. Careful site testing remains essential: a collector placed where fog is rare may gather almost nothing, no matter how well the mesh is designed.

STIMULUS END

TACHS-R4-018

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S05

Prompt:

What causes collected droplets to move from the mesh into the channel?

- A) Heat
- B) Gravity
- C) A fuel-powered pump
- D) A protective coating

TACHS-R4-019

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S05

Prompt:

Why does the author state that fog collectors “do not create water”?

- A) To explain why the mesh must be replaced often
- B) To compare the collectors with rain gauges
- C) To argue that collectors should be used only for research
- D) To clarify a limitation and prevent an exaggerated impression of the technology

TACHS-R4-020

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R4-S05

Prompt:

How is the passage mainly organized?

- A) It explains how a device works, identifies factors affecting it, and discusses its practical limits.
- B) It lists several inventions in the order they were developed.

- C) It contrasts two experiments and explains why one failed.
- D) It gives step-by-step directions for building a collector at home.

TACHS-R4-021

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S05

Prompt:

As used in the final paragraph, “suitable” most nearly means

- A) inexpensive.
- B) permanent.
- C) appropriate for the purpose.
- D) easy to reach.

STIMULUS START

Stimulus ID: TACHS-R4-S06

Section: Reading

Stimulus Type: Informational Historical-Style Passage

Title: Messages Across the Mountain

Text:

Before electronic communication reached many mountain communities, sending an urgent message could require a rider to travel for hours. In one region, a group of towns created a chain of signal stations on high ridges. The system described here is a historical-style example rather than a record of one specific network.

Each station could see the station before it and the one after it. Operators used pairs of movable wooden arms mounted on a tall post. Different arm positions represented letters, numbers, or short instructions. An operator viewed an incoming signal through a telescope, recorded it, and then repeated it for the next station. In clear weather, a message could cross the region far faster than a horse could carry it.

Speed did not guarantee accuracy. A tired operator might copy an angle incorrectly, and mist could hide a station entirely. To reduce mistakes, operators paused after a group of symbols and waited for a confirmation signal. Important messages were also written in a log so that the sequence could be checked later.

The network required trained workers and unobstructed views, which made it expensive to maintain. Eventually, telegraph wires offered a more dependable way to send messages in poor visibility and after dark. Even so, the ridge stations demonstrated an important principle: information can travel quickly when a long journey is divided among connected points.

STIMULUS END

TACHS-R4-022

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S06

Prompt:

According to the passage, what did an operator do after viewing an incoming signal?

- A) Rode to the next station with a written copy
- B) Changed the meaning into a shorter instruction
- C) Waited until darkness before forwarding it
- D) Recorded the signal and repeated it for the next station

TACHS-R4-023

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S06

Prompt:

The author's tone toward the signal system is best described as

- A) amused by its unnecessary complexity.
- B) appreciative of its ingenuity while aware of its weaknesses.
- C) doubtful that such a system could transmit any information.
- D) critical of the towns for replacing riders.

TACHS-R4-024

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S06

Prompt:

Which statement best summarizes the passage?

- A) A chain of visual stations sped communication by relaying coded signals, although weather and human error limited the system.
- B) Mountain towns used telescopes mainly to predict dangerous weather for travelers.
- C) Telegraph wires were developed because horses could not cross mountain roads.
- D) Operators preferred written logs because arm signals represented only numbers.

TACHS-R4-025

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S06

Prompt:

Why was the confirmation signal especially useful?

- A) It showed operators how to repair the wooden arms.
- B) It allowed stations to work after dark.
- C) It helped detect an incorrectly received group of symbols before the message continued.
- D) It warned riders that a message was urgent.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE4-S03

Section: Written Expression

Stimulus Type: Revision Passage

Title: Mapping Quiet Places

Text:

[1] Our student council wanted to identify calm places where students could read or work during lunch. [2] First, volunteers measured noise levels in six areas on three different days. [3] They also counted available seats and noted whether each location was sheltered from bad weather. [4] The library corner was consistently quiet, but it had only eight seats. [5] By contrast, the covered courtyard offered thirty seats and was only slightly louder. [6] Several council members enjoy mystery novels. [7] After reviewing the results, the council recommended adding two small tables to the library corner and posting quiet-zone signs in one section of the courtyard. [8] These changes would create more useful choices without requiring the school to build a new room.

STIMULUS END

TACHS-WE4-014

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S03

Prompt:

Which sentence should be deleted because it does not support the passage's central idea?

A) Sentence 6

B) Sentence 3

C) Sentence 5

D) Sentence 8

TACHS-WE4-015

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S03

Prompt:

Which revision of sentence 2 is most precise?

A) First, volunteers did some tests in a number of places.

B) First, volunteers used sound meters to record noise in six areas on three separate days.

C) First, six areas were things that volunteers looked at repeatedly.

D) First, volunteers carefully handled an interesting school concern.

TACHS-WE4-016

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE4-S03

Prompt:

Which revision of sentence 7 uses verb tense correctly?

A) After reviewing the results, the council recommends adding two small tables and posted quiet-zone signs.

B) After reviewing the results, the council had recommend adding two small tables and posts quiet-zone signs.

C) After reviewing the results, the council recommended adding two small tables and posting quiet-zone signs in one section of the courtyard.

D) After reviewing the results, the council recommending two small tables and had posted quiet-zone signs.

TACHS-WE4-017

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S03

Prompt:

Which revision most effectively combines sentences 4 and 5?

A) The library corner was consistently quiet, it had only eight seats, and the covered courtyard had thirty.

B) The library corner was quiet with eight seats, the courtyard was covered and only slightly louder with thirty.

C) Having only eight seats in the quiet library corner and thirty in the courtyard, which was covered.

D) Although the library corner was consistently quieter, the covered courtyard offered thirty seats instead of only eight.

STIMULUS START

Stimulus ID: TACHS-WE4-S04

Section: Written Expression

Stimulus Type: Revision Passage

Title: Restoring a Stream Bank

Text:

[1] Heavy rain had washed soil from the bank of a stream near our community center. [2] A local conservation group invited students to help stabilize the damaged area. [3] Before planting, the leader explained that native grasses have dense roots that hold soil in place. [4] We carried trays of seedlings down the path and arranged them in marked rows. [5] The work was muddy, and everyone stayed focused until each plant was secure. [6] Next spring, the group will return to measure how much of the bank remains covered. [7] This follow-up matters because restoration is a process, not a single afternoon of work. [8] The project showed us that careful observation after planting is as important as the planting itself.

STIMULUS END

TACHS-WE4-018

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S04

Prompt:

Which opening sentence would best introduce the passage?

A) Outdoor projects can involve many different tools and supplies.

B) Streams are bodies of moving water found in many regions.

C) One rainy Saturday, our class learned how planting and later monitoring can protect an eroded stream bank.

D) The community center offers several activities for students.

TACHS-WE4-019

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S04

Prompt:

Which revision of sentence 5 best shows the relationship between its ideas?

- A) Although the work was muddy, everyone stayed focused until each plant was secure.
- B) The work was muddy, or everyone stayed focused until each plant was secure.
- C) The work was muddy, everyone, however, stayed focused until each plant was secure.
- D) The work was muddy, everyone stayed focused until each plant was secure.

TACHS-WE4-020

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S04

Prompt:

Which revision of sentence 6 is most concise without changing its meaning?

- A) Next spring, the group will return back again in order to measure the coverage that remains on the bank.
- B) Next spring, the group will return to measure the bank's remaining plant cover.
- C) In the springtime that comes next, the group will be returning for the purpose of measurement.
- D) The group, next spring, will go back to the bank, and the amount covered will be measured by them.

TACHS-WE4-021

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S04

Prompt:

Which sentence would provide the most relevant support after sentence 3?

- A) The selected grasses also tolerate the stream bank's usual moisture and sunlight.
- B) The leader had worked with several volunteer groups before ours.
- C) Native plants can have green, brown, or yellow leaves.
- D) We wore old shoes because the path was muddy.

TACHS-WE4-022

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Neither the captain nor the players was prepared for the schedule change.

- B) Neither the players nor the captain were prepared for the schedule change.
C) Neither the captain nor the players were prepared for the schedule change.
D) Neither the players or the captain was prepared for the schedule change.

TACHS-WE4-023

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses pronouns correctly?

- A) The science teacher asked Maya and I to organize the samples.
B) The science teacher asked Maya and myself to organize the samples.
C) The science teacher asked I and Maya to organize the samples.
D) The science teacher asked Maya and me to organize the samples.

TACHS-WE4-024

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: By the time the doors opened, the volunteers had arranged the chairs.

Line 2: Each row was checked to ensure that the aisles remained clear.

Line 3: The coordinator then give every volunteer a final assignment.

Line 4: At noon, the first visitors entered the hall.

- A) Line 1
B) Line 2
C) Line 3
D) Line 4

TACHS-WE4-025

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The two proposals address the same problem.

Line 2: Of the two plans, Marisol's is the most practical.

Line 3: Her plan requires fewer materials and less time.

Line 4: The committee will discuss both options tomorrow.

- A) Line 1
B) Line 2
C) Line 3
D) Line 4

TACHS-WE4-026

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: Luis placed the glass bowl carefully on the counter.

Line 2: He checked the recipe before adding the flour.

Line 3: The batter became smooth after several minutes of mixing.

Line 4: He poured it slow into the prepared pan.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M4-013

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A recipe uses $2\frac{1}{4}$ cups of oats for 18 bars. At the same rate, how many cups of oats are needed for 30 bars?

A) $2\frac{3}{4}$ cups

B) 3 cups

C) $3\frac{1}{4}$ cups

D) $3\frac{3}{4}$ cups

TACHS-M4-014

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the greatest common factor of 84 and 126?

A) 42

B) 21

C) 14

D) 7

TACHS-M4-015

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

At sunrise, the temperature was -6°C . By noon, it had risen 11°C , and by evening, it had fallen 8°C . What was the evening temperature?

- A) 13°C
- B) -3°C
- C) 3°C
- D) -13°C

TACHS-M4-016

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

For the function $y = 3x - 4$, which ordered pair is a solution?

- A) (1, 1)
- B) (2, 4)
- C) (3, 5)
- D) (4, 7)

TACHS-M4-017

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $5(x - 2) = 3x + 8$.

- A) $x = 7$
- B) $x = 8$
- C) $x = 10$
- D) $x = 9$

TACHS-M4-018

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of minutes four students read on Monday were 18, 24, 31, and 37. On Tuesday, each student read exactly 5 more minutes than on Monday. How does the Tuesday data set compare with the Monday data set?

- A) The mean increases by 5, and the range increases by 5.
- B) The mean increases by 5, and the range stays the same.
- C) The median stays the same, and the range increases by 5.
- D) Both the mean and the range stay the same.

TACHS-M4-019

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A spinner has eight equal sections numbered 1 through 8. What is the probability of spinning a number that is both even and greater than 3?

A) $\frac{3}{8}$

B) $\frac{1}{4}$

C) $\frac{1}{2}$

D) $\frac{5}{8}$

TACHS-M4-020

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A club recorded the number of books collected during four weeks.

Week 1: 36

Week 2: 48

Week 3: 42

Week 4: 54

What percent of the total books were collected during Week 4?

A) 20%

B) 25%

C) 27%

D) 30%

TACHS-M4-021

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A right triangle has legs of 9 centimeters and 12 centimeters. What is the length of its hypotenuse?

A) 18 cm

B) 21 cm

C) 15 cm

D) 144 cm

TACHS-M4-022

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A circle has a diameter of 14 inches. Using $\frac{22}{7}$ for π , what is the circumference?

A) 44 inches

- B) 22 inches
- C) 88 inches
- D) 154 inches

TACHS-M4-023

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A square has an area of 81 square units. A second square has side lengths twice as long. What is the area of the second square?

- A) 162 square units
- B) 324 square units
- C) 243 square units
- D) 648 square units

TACHS-M4-024

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A map scale states that 1 inch represents 18 miles. Two towns are $3\frac{1}{2}$ inches apart on the map. What is the actual distance between them?

- A) 54 miles
- B) 72 miles
- C) 63 miles
- D) 81 miles

TACHS-M4-025

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A rectangular storage box is 10 inches long, 6 inches wide, and 4 inches high. How many 1-inch cubes can completely fill the box?

- A) 20
- B) 120
- C) 200
- D) 240

SECTION 4 - ABILITY

TACHS-A4-013

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ○ above -

Top-right: - with ○ below

Bottom-left: □ on the left of |

Bottom-right: ?

Across each row, the shape moves to the opposite side of the line. Down each column, ○ changes to □ and - rotates to |.

A) | with □ on the left

B) | with □ on the right

C) - with □ below

D) | with ○ on the right

TACHS-A4-014

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ○ | □ | △

Row 2: □ | △ | ○

Row 3: △ | ○ | ?

Each row and column contains one circle, one square, and one triangle.

A) ○

B) □

C) ◇

D) △

TACHS-A4-015

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Open shapes: ○ □ △

Filled shapes: ● ■ ▲

Complete the 2×2 matrix.

Top-left: ○□

Top-right: ●□
Bottom-left: ○△
Bottom-right: ?

Across each row, only the first shape changes from open to filled. Down each column, only the second shape changes from square to triangle.

- A) ●△
- B) ▲○
- C) ○▲
- D) ●□

TACHS-A4-016

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ↑ | → | ↓
Row 2: → | ↓ | ←
Row 3: ↓ | ← | ?

Across each row and down each column, the arrow rotates 90° clockwise from one cell to the next.

- A) ←
- B) ↓
- C) ↑
- D) →

TACHS-A4-017

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded in half from bottom to top. One hole is punched near the lower-left corner of the folded rectangle, away from the fold. When the sheet is unfolded, which pattern appears?

- A) Two holes near the upper-left and lower-left corners
- B) Two holes near the upper-left and upper-right corners
- C) Four holes, one near each corner
- D) Two holes near the lower-left and lower-right corners

TACHS-A4-018

Section: Ability

Type: mcq

Category: Ability
Skill: Paper Folding
Stimulus ID: None

Prompt:

A square sheet is folded along the diagonal from upper-right to lower-left. A hole is punched near the lower-left corner of the folded triangle, away from the fold. What appears when the sheet is unfolded?

- A) Four holes symmetric across both diagonals
- B) One hole on the diagonal
- C) Two holes mirrored across the upper-right-to-lower-left diagonal
- D) Two holes mirrored across the horizontal centerline

TACHS-A4-019

Section: Ability

Type: mcq

Category: Ability
Skill: Paper Folding
Stimulus ID: None

Prompt:

A square sheet is folded from left to right and then from bottom to top. A hole is punched at the exact center of the final folded square, away from both folded edges. What pattern appears after complete unfolding?

- A) Two holes on the vertical centerline
- B) Eight holes around the border
- C) One hole at the center of the original sheet
- D) Four holes, one in each quadrant, symmetric across both centerlines

TACHS-A4-020

Section: Ability

Type: mcq

Category: Ability
Skill: Paper Folding
Stimulus ID: None

Prompt:

A rectangular sheet is folded in half from right to left. A small semicircular cut is made at the midpoint of the folded rectangle's top outer edge, away from the fold. What appears along the top edge when the sheet is unfolded?

- A) One circular opening at the center
- B) Two matching semicircular notches, one in each half of the top edge
- C) Four semicircular notches along the top edge
- D) One semicircular notch at the left corner

TACHS-A4-021

Section: Ability

Type: mcq

Category: Ability
Skill: Figure Classification
Stimulus ID: None

Prompt:

The three arrangements below share the same rule.

Arrangement 1: ○ □ ○ □

Arrangement 2: △ ◇ △ ◇

Arrangement 3: ☆ + ☆ +

Which choice follows the same rule?

A) ○ ○ □ □

B) ▲ ▲ ▲ ▲

C) ◆ □ ◆ □

D) + × + ×

TACHS-A4-022

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three figures below share the same rule.

Figure 1: Top: ○; Center: □; Bottom: ○

Figure 2: Top: △; Center: ◇; Bottom: △

Figure 3: Top: ☆; Center: +; Bottom: ☆

Which choice follows the same rule?

A) Top: □; Center: ○; Bottom: □

B) Top: ○; Center: □; Bottom: △

C) Top: ●; Center: □; Bottom: ○

D) Top: +; Center: +; Bottom: ×

TACHS-A4-023

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three arrangements below share the same rule.

Arrangement 1: ○ inside □; ● outside □

Arrangement 2: △ inside ◇; ▲ outside ◇

Arrangement 3: ☆ inside ○; ★ outside ○

Which choice follows the same rule?

A) ■ inside □; □ outside □

B) ◆ inside △; ◇ outside △

C) ◇ inside ☆; ◆ outside ☆

D) ▲ inside ○; ▲ outside ○

TACHS-A4-024

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three sequences below share the same rule.

Sequence 1: $\uparrow \uparrow \rightarrow$

Sequence 2: $\rightarrow \rightarrow \downarrow$

Sequence 3: $\downarrow \downarrow \leftarrow$

Which choice follows the same rule?

A) $\leftarrow \uparrow \uparrow$

B) $\uparrow \rightarrow \rightarrow$

C) $\leftarrow \downarrow \leftarrow$

D) $\leftarrow \leftarrow \uparrow$

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R4-014 - Correct: A - Correct Answer: Pale marks also appeared on objects fired near the kiln door. - Explanation: Both test tiles from the same area show marks like those on Mara's bowl.

TACHS-R4-015 - Correct: B - Correct Answer: He suspects its position may help explain the streaks. - Explanation: Comparing nearby test tiles helps connect the uneven glaze to cool air entering by the door.

TACHS-R4-016 - Correct: C - Correct Answer: Examining an unexpected result can turn a flaw into useful information. - Explanation: Mara and Mr. Chen use the streaks to identify and correct a faulty kiln seal.

TACHS-R4-017 - Correct: D - Correct Answer: information that helps support an explanation. - Explanation: The uneven glaze supplies information supporting the explanation that the kiln door seal was worn.

TACHS-R4-018 - Correct: B - Correct Answer: Gravity - Explanation: The passage states that gravity pulls drops heavy enough to move into the channel.

TACHS-R4-019 - Correct: D - Correct Answer: To clarify a limitation and prevent an exaggerated impression of the technology - Explanation: The statement emphasizes that collectors capture existing moisture and cannot provide unlimited water.

TACHS-R4-020 - Correct: A - Correct Answer: It explains how a device works, identifies factors affecting it, and discusses its practical limits. - Explanation: The paragraphs move from operation to design conditions and then to uses and limitations.

TACHS-R4-021 - Correct: C - Correct Answer: appropriate for the purpose. - Explanation: Suitable conditions are the fog and wind conditions appropriate for effective collection.

TACHS-R4-022 - Correct: D - Correct Answer: Recorded the signal and repeated it for the next station - Explanation: Operators wrote down incoming symbols before relaying them to the following station.

TACHS-R4-023 - Correct: B - Correct Answer: appreciative of its ingenuity while aware of its weaknesses. - Explanation: The author highlights the network's speed and principle while acknowledging weather, error, and expense.

TACHS-R4-024 - Correct: A - Correct Answer: A chain of visual stations sped communication by relaying coded signals, although weather and human error limited the system. -

Explanation: This choice includes both the system's central method and its principal limitations.

TACHS-R4-025 - Correct: C - Correct Answer: It helped detect an incorrectly received group of symbols before the message continued. - Explanation: Waiting for confirmation allowed an operator to verify each symbol group before forwarding more.

WRITTEN EXPRESSION ANSWERS

TACHS-WE4-014 - Correct: A - Correct Answer: Sentence 6 - Explanation: The council members' reading preferences do not help explain the study or its recommendations.

TACHS-WE4-015 - Correct: B - Correct Answer: First, volunteers used sound meters to record noise in six areas on three separate days. - Explanation: This revision precisely identifies the tool, measurement, locations, and schedule.

TACHS-WE4-016 - Correct: C - Correct Answer: After reviewing the results, the council recommended adding two small tables and posting quiet-zone signs in one section of the courtyard. - Explanation: The past-tense verb and parallel gerunds correctly describe the council's completed recommendation.

TACHS-WE4-017 - Correct: D - Correct Answer: Although the library corner was consistently quieter, the covered courtyard offered thirty seats instead of only eight. - Explanation: The subordinating conjunction clearly expresses contrast without creating a fragment or run-on.

TACHS-WE4-018 - Correct: C - Correct Answer: One rainy Saturday, our class learned how planting and later monitoring can protect an eroded stream bank. - Explanation: The sentence introduces the setting and both major stages developed in the passage.

TACHS-WE4-019 - Correct: A - Correct Answer: Although the work was muddy, everyone stayed focused until each plant was secure. - Explanation: "Although" correctly shows contrast and joins the ideas in one complete sentence.

TACHS-WE4-020 - Correct: B - Correct Answer: Next spring, the group will return to measure the bank's remaining plant cover. - Explanation: This version preserves the meaning while eliminating wordiness and repetition.

TACHS-WE4-021 - Correct: A - Correct Answer: The selected grasses also tolerate the stream bank's usual moisture and sunlight. - Explanation: This detail directly supports why those native grasses were appropriate for the site.

TACHS-WE4-022 - Correct: C - Correct Answer: Neither the captain nor the players were prepared for the schedule change. - Explanation: With "neither...nor," the verb agrees with the nearer plural subject "players."

TACHS-WE4-023 - Correct: D - Correct Answer: The science teacher asked Maya and me to organize the samples. - Explanation: "Me" is the correct objective pronoun after the verb "asked."

TACHS-WE4-024 - Correct: C - Correct Answer: Line 3 - Explanation: The past-tense context requires "gave," not the present-tense form "give."

TACHS-WE4-025 - Correct: B - Correct Answer: Line 2 - Explanation: A comparison between two plans requires the comparative "more practical," not the superlative "most practical."

TACHS-WE4-026 - Correct: D - Correct Answer: Line 4 - Explanation: The adverb "slowly," rather than the adjective "slow," must modify "poured."

MATHEMATICS ANSWERS

TACHS-M4-013 - Correct: D - Correct Answer: $3\frac{3}{4}$ cups - Explanation: $2\frac{1}{4} \div 18 = \frac{1}{8}$ cup per bar, and $30 \times \frac{1}{8} = 3\frac{3}{4}$ cups.

TACHS-M4-014 - Correct: A - Correct Answer: 42 - Explanation: The common factors of 84 and 126 include 42, and no larger number divides both.

TACHS-M4-015 - Correct: B - Correct Answer: -3°C - Explanation: Starting at -6 , the changes give $-6 + 11 - 8 = -3$.

TACHS-M4-016 - Correct: C - Correct Answer: (3, 5) - Explanation: Substituting $x = 3$ gives $y = 3(3) - 4 = 5$.

TACHS-M4-017 - Correct: D - Correct Answer: $x = 9$ - Explanation: Expanding gives $5x - 10 = 3x + 8$, so $2x = 18$ and $x = 9$.

TACHS-M4-018 - Correct: B - Correct Answer: The mean increases by 5, and the range stays the same. - Explanation: Adding 5 to every value adds 5 to the mean but does not change the difference between extremes.

TACHS-M4-019 - Correct: A - Correct Answer: $3/8$ - Explanation: The favorable outcomes are 4, 6, and 8, giving 3 favorable sections out of 8.

TACHS-M4-020 - Correct: D - Correct Answer: 30% - Explanation: The total is 180 books, and $54 \div 180 = 0.30 = 30\%$.

TACHS-M4-021 - Correct: C - Correct Answer: 15 cm - Explanation: The hypotenuse is $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$ centimeters.

TACHS-M4-022 - Correct: A - Correct Answer: 44 inches - Explanation: Circumference equals πd , so $22/7 \times 14 = 44$ inches.

TACHS-M4-023 - Correct: B - Correct Answer: 324 square units - Explanation: The first side is 9; doubling it gives 18, and $18^2 = 324$.

TACHS-M4-024 - Correct: C - Correct Answer: 63 miles - Explanation: The actual distance is $3 \frac{1}{2} \times 18 = 63$ miles.

TACHS-M4-025 - Correct: D - Correct Answer: 240 - Explanation: The box's volume is $10 \times 6 \times 4 = 240$ cubic inches, so it holds 240 unit cubes.

ABILITY ANSWERS

TACHS-A4-013 - Correct: B - Correct Answer: | with $\square$ on the right - Explanation: The shape crosses to the opposite side, while moving down changes the shape and rotates the line.

TACHS-A4-014 - Correct: B - Correct Answer: $\square$ - Explanation: A square completes the set of three different shapes in both the final row and final column.

TACHS-A4-015 - Correct: A - Correct Answer: $\bullet\Delta$ - Explanation: The first shape becomes filled across the row, and the second changes from square to triangle down the column.

TACHS-A4-016 - Correct: C - Correct Answer: $\uparrow$ - Explanation: Continuing the 90° clockwise rotation after the left arrow produces an up arrow.

TACHS-A4-017 - Correct: A - Correct Answer: Two holes near the upper-left and lower-left corners - Explanation: Unfolding across the horizontal centerline reflects the punched hole vertically on the left side.

TACHS-A4-018 - Correct: C - Correct Answer: Two holes mirrored across the upper-right-to-lower-left diagonal - Explanation: Reversing the diagonal fold reflects the hole once across that same diagonal.

TACHS-A4-019 - Correct: D - Correct Answer: Four holes, one in each quadrant, symmetric across both centerlines - Explanation: Two perpendicular folds create four reflected locations, one in each quadrant.

TACHS-A4-020 - Correct: B - Correct Answer: Two matching semicircular notches, one in each half of the top edge - Explanation: Unfolding reflects the outer-edge cut to the corresponding position on the other half.

TACHS-A4-021 - Correct: D - Correct Answer: $+ \times + \times$ - Explanation: Each valid arrangement alternates two distinct open symbols in an ABAB pattern.

TACHS-A4-022 - Correct: A - Correct Answer: Top: $\square$; Center: $\bigcirc$; Bottom: $\square$ - Explanation: The top and bottom symbols match, while the center symbol is different.

TACHS-A4-023 - Correct: C - Correct Answer: ◇ inside ☆; ◆ outside ☆ - Explanation: Each group places an open shape inside a boundary and its filled counterpart outside.

TACHS-A4-024 - Correct: D - Correct Answer: ← ← ↑ - Explanation: The first direction repeats, and the third arrow is its 90° clockwise rotation.

TACHS Practice Exam - Exam 4 - Part D
Filename: tachs_full_exam_04_part_d.txt
Exam Family: TACHS
Exam Number: 4
Exam Part: D
Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R4-S10

Section: Reading

Stimulus Type: Literary or Narrative Passage

Title: The Stubborn Pulley

Text:

Nico was helping the stage crew raise a painted backdrop for the school play. During rehearsal, the backdrop stopped halfway up and tilted toward one side. Pulling harder only made the rope groan.

Ms. Patel asked everyone to step back. She traced the rope from the handle to the ceiling and noticed that one pulley wheel barely turned. Dust had collected around its axle, and the rope had slipped toward the rim. Nico suggested spraying oil from the floor, but Ms. Patel explained that too much oil could drip onto the fabric and attract more dust.

After the ladder was secured, she cleaned the axle with a narrow brush and applied one small drop of lubricant. Nico guided the rope into the center groove while another student kept the backdrop supported. They tested the system with a sandbag before attaching the painted cloth again.

This time the backdrop rose evenly. Nico had expected the solution to require a stronger pull or a new rope. Instead, the repair depended on reducing resistance at one small point. During the performance, the audience applauded the scenery change. The crew listened for the pulley and heard almost nothing.

STIMULUS END

TACHS-R4-039

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S10

Prompt:

What problem does Ms. Patel find in the pulley system?

- A) The painted cloth is too heavy for the rope.
- B) The handle is attached to the wrong wall.
- C) One pulley wheel barely turns and the rope has shifted toward its rim.
- D) A sandbag is caught above the backdrop.

TACHS-R4-040

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S10

Prompt:

Why does the crew test the pulley with a sandbag before reattaching the backdrop?

- A) To confirm that the repair works without risking the painted cloth
- B) To make the rope stretch before the performance
- C) To clean dust from the ceiling
- D) To show the actors how scenery is moved

TACHS-R4-041

Section: Reading

Type: mcq
Category: Reading
Skill: Main Idea, Central Thought, or Summary
Stimulus ID: TACHS-R4-S10
Prompt:
Which statement best expresses a central idea of the passage?
A) Stage scenery should always be moved by adults.
B) A new rope solves most pulley problems.
C) Audiences notice every backstage repair.
D) A small source of resistance can matter more than additional force.

TACHS-R4-042
Section: Reading
Type: mcq
Category: Reading
Skill: Vocabulary in Context
Stimulus ID: TACHS-R4-S10
Prompt:
As used in the third paragraph, "secured" most nearly means
A) decorated.
B) made stable and safe.
C) hidden from view.
D) measured for height.

STIMULUS START
Stimulus ID: TACHS-R4-S11
Section: Reading
Stimulus Type: Informational or Scientific Passage
Title: How Antlion Pits Trap Moving Sand
Text:
An antlion larva lives beneath loose, dry sand and builds a cone-shaped pit. It begins by moving backward in a circle while flicking grains away with its head. As the circle tightens, the center becomes deeper and the sides become steep.

The slope of the finished pit is close to the steepest angle at which the sand can remain stable. When a small insect steps onto the edge, grains begin to slide. The insect tries to climb, but each movement may loosen more sand. The antlion waits below the center and sometimes throws additional grains upward, creating more small slides.

The pit does not pull an insect downward like a machine. Instead, it uses the behavior of loose grains. A shallow slope would be easier to cross, while a much steeper slope would collapse before the trap was ready. The antlion repeatedly repairs the sides so they remain near the useful angle.

Scientists study these pits to understand movement on unstable surfaces. The same principles can help explain why feet, wheels, and digging tools behave differently in sand than on solid ground. The antlion's trap is effective because its shape works with the material rather than against it.
STIMULUS END

TACHS-R4-043
Section: Reading
Type: mcq
Category: Reading
Skill: Factual Comprehension, Detail, or Textual Evidence
Stimulus ID: TACHS-R4-S11
Prompt:
How does an antlion begin building its pit?
A) By carrying wet sand toward the center
B) By moving backward in a circle and flicking grains away
C) By digging a straight tunnel beneath a stone
D) By pressing the sand into a hard wall

TACHS-R4-044
Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S11

Prompt:

Why does the author explain that the pit does not pull an insect downward like a machine?

- A) To show that antlions are too weak to move sand
- B) To compare the pit with a mechanical elevator
- C) To argue that machines cannot work on loose ground
- D) To emphasize that the trap depends on the behavior of unstable grains

TACHS-R4-045

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R4-S11

Prompt:

How does the third paragraph contribute to the passage?

- A) It explains why the pit must have a carefully balanced slope.
- B) It lists several animals that escape antlion pits.
- C) It gives instructions for keeping an antlion as a pet.
- D) It describes how rain fills the pit with water.

TACHS-R4-046

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R4-S11

Prompt:

As used in the final paragraph, "principles" most nearly means

- A) official rules for protecting insects.
- B) tools used to measure the pit.
- C) basic ideas that explain how something works.
- D) steps performed in a fixed order.

STIMULUS START

Stimulus ID: TACHS-R4-S12

Section: Reading

Stimulus Type: Humanities or Reflective Passage

Title: Why Translators Keep More Than One Draft

Text:

Translating a paragraph involves more than replacing each word with a word from another language. A sentence may contain an expression that sounds natural to one audience but confusing to another. Its rhythm may also affect whether the speaker sounds formal, playful, impatient, or calm.

For this reason, translators often prepare several drafts. One draft may stay close to the original sentence structure so that no idea is lost. A second may rearrange the wording to sound natural in the new language. A third may focus on tone, especially when the text includes humor, dialogue, or instructions.

The drafts are then compared. A smooth sentence is not successful if it changes the meaning, and a perfectly literal sentence may fail if readers cannot understand its purpose. Translators may read a passage aloud, ask another speaker to review it, or return to surrounding paragraphs for context.

The final version hides most of this comparison. Readers see one set of words, not the abandoned choices beside it. Yet those alternatives are valuable. They allow the translator to test meaning, clarity, and voice separately before selecting language that balances all three.

STIMULUS END

TACHS-R4-047

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R4-S12

Prompt:

What may a translator's second draft do according to the passage?

- A) Remove every difficult sentence
- B) Copy the original word order exactly
- C) Add facts not included in the original
- D) Rearrange wording so it sounds natural in the new language

TACHS-R4-048

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R4-S12

Prompt:

Why might a translator read a passage aloud?

- A) To replace the written version with a recording
- B) To judge how naturally the wording and tone sound
- C) To avoid comparing the passage with surrounding paragraphs
- D) To count the number of words in each sentence

TACHS-R4-049

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R4-S12

Prompt:

What does the author emphasize by saying that the final version "hides" the comparison?

- A) Translators should never share early drafts.
- B) Readers are unable to understand translated writing.
- C) A finished translation may conceal extensive testing of alternatives.
- D) Only the shortest draft should be published.

TACHS-R4-050

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R4-S12

Prompt:

Which statement best summarizes the passage?

- A) Multiple drafts help translators balance meaning, clarity, and tone.
- B) Literal translations are always more accurate than natural ones.
- C) Every translated paragraph should preserve the original word order.
- D) Translators mainly revise a text to make it shorter.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE4-S07

Section: Written Expression

Stimulus Type: Revision Passage

Title: Improving Locker-Number Visibility

Text:

[1] Students reported that locker numbers were difficult to read from the center of the hallway. [2] Our design class photographed the lockers from several distances. [3] We compared the existing silver numbers with paper samples in dark blue, black, and white. [4] The black samples produced the strongest contrast against the pale doors. [5] We also tested two number sizes during a passing period. [6] More students identified the larger numbers correctly while walking. [7] The hallway contains eighty-four lockers. [8] We recommended larger black numbers placed at the same

height on every door.
STIMULUS END

TACHS-WE4-039

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S07

Prompt:

Which sentence would best follow sentence 2?

- A) Some students prefer lockers near the stairwell.
- B) The photographs showed that glare and low contrast hid many numbers.
- C) The class camera had a rechargeable battery.
- D) The hallway floor was polished every Friday.

TACHS-WE4-040

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S07

Prompt:

Which revision of sentence 5 is most effective?

- A) During a passing period, we also tested two number sizes.
- B) Two number sizes during a passing period, and we tested them.
- C) During a passing period two number sizes were, they were tested.
- D) Testing two number sizes during a passing period, the students.

TACHS-WE4-041

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S07

Prompt:

Which revision of sentence 6 is clearest?

- A) While walking through the hall, students correctly read the larger numbers more often.
- B) The larger numbers were better and got more correct.
- C) Walking, the numbers were identified by more students correctly.
- D) More students did it right when the numbers got bigger.

TACHS-WE4-042

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE4-S07

Prompt:

Which sentence should be deleted because it is not needed to develop the passage?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 7
- D) Sentence 8

STIMULUS START

Stimulus ID: TACHS-WE4-S08

Section: Written Expression

Stimulus Type: Revision Passage

Title: Building a Miniature Wind Tunnel

Text:

[1] Our science group built a small wind tunnel from a clear storage box and a quiet fan. [2] A cardboard grid at the entrance straightened the moving air. [3] We suspended paper shapes from a thin rod inside the box. [4] Each shape was tested at the same fan setting for thirty seconds. [5]

The narrow triangle moved less than the wide rectangle. [6] Because the shapes had different areas, we could not claim that outline alone caused the difference. [7] We plan to repeat the test with shapes cut to equal areas. [8] The first trial helped us design a fairer second experiment.
STIMULUS END

TACHS-WE4-043

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE4-S08

Prompt:

Which revision most effectively combines sentences 2 and 3?

- A) After a cardboard grid straightened the moving air, we suspended paper shapes from a thin rod inside the box.
- B) A cardboard grid straightened the air, the paper shapes were suspended from a rod.
- C) Straightening the air, the paper shapes suspended from a rod.
- D) The grid and the shapes, which were in the box and straightened.

TACHS-WE4-044

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE4-S08

Prompt:

Which revision of sentence 5 is most precise?

- A) The triangle did better than the rectangle.
- B) One shape moved in a different way from another shape.
- C) At the same fan setting, the narrow triangle swung through a smaller distance than the wide rectangle.
- D) The shapes acted like shapes in moving air.

TACHS-WE4-045

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE4-S08

Prompt:

Which revision of sentence 7 uses standard written US English?

- A) We plans to repeat the test with equal areas.
- B) We plan to repeat the test with shapes cut to equal areas.
- C) We planning to repeat the test with shapes of equal area.
- D) We plan repeating the test with equal areas yesterday.

TACHS-WE4-046

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The boxes that was delivered today are in the office.
- B) The boxes that were delivered today are in the office.
- C) The boxes who were delivered today is in the office.
- D) The boxes delivered today was in the office.

TACHS-WE4-047

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses the modifier correctly?

- A) Walking into the lab, the experiment was already running.
- B) Covered in labels, Maya organized the storage shelf.
- C) Walking into the lab, Maya noticed that the experiment was already running.
- D) After reading the directions, the machine was started.

TACHS-WE4-048

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The model rested on a wooden base.

Line 2: Two wires connected it to the battery.

Line 3: The students tested the switch twice.

Line 4: Each of the lights were visible from the doorway.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE4-049

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The guide spoke clearly to the group.

Line 2: Everyone followed her into the workshop.

Line 3: She explained the safety rules first.

Line 4: Then she demonstrated the tool careful.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE4-050

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The team finished its first draft.

Line 2: Two paragraphs needed additional evidence.

Line 3: The writers revised those sections.

Line 4: The editors comments were added in blue.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M4-038

Section: Mathematics

Type: mcq

Category: Mathematics
Skill: Number Sense and Operations
Stimulus ID: None
Prompt:
Which fraction is exactly halfway between $\frac{2}{5}$ and $\frac{3}{5}$?
A) $\frac{3}{8}$
B) $\frac{9}{20}$
C) $\frac{1}{2}$
D) $\frac{11}{20}$

TACHS-M4-039
Section: Mathematics
Type: mcq
Category: Mathematics
Skill: Number Sense and Operations
Stimulus ID: None
Prompt:
Which number is equal to $2^3 \times 3^2$?
A) 36
B) 48
C) 64
D) 72

TACHS-M4-040
Section: Mathematics
Type: mcq
Category: Mathematics
Skill: Number Sense and Operations
Stimulus ID: None
Prompt:
What is the smallest whole number that must be added to 587 to make the result divisible by 9?
A) 7
B) 5
C) 4
D) 2

TACHS-M4-041
Section: Mathematics
Type: mcq
Category: Mathematics
Skill: Algebraic Patterns and Connections
Stimulus ID: None
Prompt:
Which ordered pair satisfies both $y = -2x + 5$ and $x + y = 2$?
A) (1, 1)
B) (3, -1)
C) (-1, 3)
D) (2, 0)

TACHS-M4-042
Section: Mathematics
Type: mcq
Category: Mathematics
Skill: Algebraic Patterns and Connections
Stimulus ID: None
Prompt:
A line has a slope of -3 and passes through (2, 1). What is the y-coordinate when $x = 5$?
A) -14
B) -10
C) -8
D) 8

TACHS-M4-043
Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The data set is 4, 5, 6, 7, 8. If 30 is added, which measure changes more: the mean or the median?

- A) The mean changes more.
- B) The median changes more.
- C) They change by the same amount.
- D) Neither measure changes.

TACHS-M4-044

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A whole number from 1 through 20 is selected at random. What is the probability that it is divisible by 3 or by 5?

- A) $\frac{7}{20}$
- B) $\frac{2}{5}$
- C) $\frac{1}{2}$
- D) $\frac{9}{20}$

TACHS-M4-045

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A runner completes 3 laps in 12 minutes each and 2 laps in 18 minutes each. What is the mean time per lap?

- A) 13.2 minutes
- B) 14 minutes
- C) 14.4 minutes
- D) 15 minutes

TACHS-M4-046

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A parallelogram has an area of 84 square centimeters and a base of 12 centimeters. What is its height?

- A) 5 centimeters
- B) 7 centimeters
- C) 8 centimeters
- D) 14 centimeters

TACHS-M4-047

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangle has vertices $(-2, -1)$, $(-2, 4)$, $(5, 4)$, and $(5, -1)$. What is its area?

- A) 24 square units
- B) 28 square units
- C) 30 square units

D) 35 square units

TACHS-M4-048

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

The point (4, -3) is reflected across the y-axis. What are the coordinates of its image?

A) (-4, -3)

B) (4, 3)

C) (-3, 4)

D) (3, -4)

TACHS-M4-049

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A 3.5-pound bag of apples costs \$8.75. What is the cost per pound?

A) \$2.25

B) \$2.50

C) \$2.75

D) \$3.00

TACHS-M4-050

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

Use 1 cubic meter = 1,000 liters. A container has a volume of 0.012 cubic meter. How many liters does it hold?

A) 1.2 liters

B) 10 liters

C) 12 liters

D) 120 liters

SECTION 4 - ABILITY

TACHS-A4-038

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: $\circ \rightarrow$

Top-right: $\bullet \downarrow$

Bottom-left: $\square \uparrow$

Bottom-right: ?

Moving right fills the shape and rotates the arrow 90° clockwise.

A) $\square \rightarrow$

B) $\blacksquare \leftarrow$

C) $\square \downarrow$

D) $\blacksquare \rightarrow$

TACHS-A4-039

Section: Ability

Type: mcq
 Category: Ability
 Skill: Figure Matrices
 Stimulus ID: None
 Prompt:
 Complete the 2×3 matrix.
 Row 1: ○○ | ○ | ●●
 Row 2: △△△ | △△ | ?
 In each row, subtract the second count from the first, then fill the result.
 A) ▲
 B) △
 C) ▲▲
 D) △△

TACHS-A4-040
 Section: Ability
 Type: mcq
 Category: Ability
 Skill: Figure Matrices
 Stimulus ID: None
 Prompt:
 Complete the 3×3 matrix.
 Row 1: ○ | □ | △
 Row 2: □ | △ | ○
 Row 3: △ | ○ | ?
 Each row shifts the symbols one position left.
 A) ○
 B) △
 C) □
 D) ◇

TACHS-A4-041
 Section: Ability
 Type: mcq
 Category: Ability
 Skill: Figure Matrices
 Stimulus ID: None
 Prompt:
 Complete the 2×2 matrix.
 Top-left: Left ○; Right ●
 Top-right: Top ○; Bottom ●
 Bottom-left: Left □; Right ■
 Bottom-right: ?
 Moving right rotates the arrangement 90° clockwise.
 A) Top ■; Bottom □
 B) Top □; Bottom ■
 C) Left ■; Right □
 D) Bottom □; Top ■

TACHS-A4-042
 Section: Ability
 Type: mcq
 Category: Ability
 Skill: Paper Folding
 Stimulus ID: None
 Prompt:
 A rectangular sheet is folded from left to right, then from bottom to top. A hole is punched near the upper-left corner of the final packet, away from edges and folds. What appears after unfolding?
 A) Two holes in a horizontal row
 B) Two holes in a vertical column
 C) Three holes forming a triangle
 D) Four holes in a 2×2 arrangement

TACHS-A4-043

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded along the diagonal from lower-left to upper-right. A triangular notch is cut from the folded packet's outer bottom edge, away from the fold. What appears after unfolding?

- A) One notch on the diagonal
- B) Two matching notches reflected across the diagonal
- C) Four notches, one on each edge
- D) One diamond-shaped hole at the center

TACHS-A4-044

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A long rectangle is divided into three equal horizontal strips. The top strip is folded down onto the middle, and the bottom strip is folded up onto the middle. One hole is punched near the left side of the final stack. What appears after unfolding?

- A) Three holes in a vertical column near the left side
- B) Three holes in a horizontal row
- C) Six holes in two columns
- D) Two holes near the left edge

TACHS-A4-045

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from top to bottom. A semicircular notch is cut from the midpoint of the folded edge created by the fold. What appears after unfolding?

- A) Two notches on the top and bottom edges
- B) Two holes beside the center
- C) Four semicircular notches
- D) One circular hole centered on the horizontal midline

TACHS-A4-046

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share the same rule:

○ ● ○

□ ■ □

△ ▲ △

Which choice follows the same rule?

- A) ◆ ◇ ◆
- B) ◇ ◆ ◇
- C) ◇ ◇ ◆
- D) ◆ ◆ ◇

TACHS-A4-047

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

↑ ∞

→ □□

↓ △△

Which choice follows the same rule?

A) ← ◇◇

B) ← ◆◆

C) ↑ ◇◇

D) ← ◆◆

TACHS-A4-048

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share the same rule:

○ | ●

□ - ■

△ / ▲

Which choice follows the same rule?

A) ◇ ◆ \

B) ◆ \ ◇

C) ◇ \ ◆

D) ◇ ◆ |

TACHS-A4-049

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

↑ → ←

→ ↓ ↑

↓ ← →

Which choice follows the same rule?

A) ← ↓ ↑

B) ← → ↓

C) ← ↑ ↓

D) ↑ ← ↓

TACHS-A4-050

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share the same rule:

Open circle ○ containing ■

Open square □ containing ▲

Open diamond ◇ containing ●

Which choice follows the same rule?

A) Filled triangle ▲ containing ◆

B) Open triangle △ containing ◇

C) Filled triangle ▲ containing ○

D) Open triangle △ containing ◆

ANSWER KEY + EXPLANATIONS

READING ANSWERS

- TACHS-R4-039 - Correct: C - Correct Answer: One pulley wheel barely turns and the rope has shifted toward its rim. - Explanation: Ms. Patel observes a nearly stuck pulley wheel and a rope displaced from the center groove.
- TACHS-R4-040 - Correct: A - Correct Answer: To confirm that the repair works without risking the painted cloth - Explanation: The sandbag allows the crew to test the repaired system while protecting the backdrop.
- TACHS-R4-041 - Correct: D - Correct Answer: A small source of resistance can matter more than additional force. - Explanation: The backdrop rises after the crew reduces friction at one pulley rather than pulling harder.
- TACHS-R4-042 - Correct: B - Correct Answer: made stable and safe. - Explanation: The ladder is made stable and safe before Ms. Patel climbs it.
- TACHS-R4-043 - Correct: B - Correct Answer: By moving backward in a circle and flicking grains away - Explanation: The first paragraph says the larva circles backward while throwing grains outward.
- TACHS-R4-044 - Correct: D - Correct Answer: To emphasize that the trap depends on the behavior of unstable grains - Explanation: The clarification highlights that sliding sand, not a mechanical pulling force, makes the trap effective.
- TACHS-R4-045 - Correct: A - Correct Answer: It explains why the pit must have a carefully balanced slope. - Explanation: The paragraph contrasts slopes that are too shallow or too steep and explains the useful balance.
- TACHS-R4-046 - Correct: C - Correct Answer: basic ideas that explain how something works. - Explanation: The word refers to general ideas about movement on unstable materials.
- TACHS-R4-047 - Correct: D - Correct Answer: Rearrange wording so it sounds natural in the new language - Explanation: The passage says a second draft may reorganize wording to sound natural.
- TACHS-R4-048 - Correct: B - Correct Answer: To judge how naturally the wording and tone sound - Explanation: Reading aloud helps a translator evaluate flow, voice, and clarity in the new language.
- TACHS-R4-049 - Correct: C - Correct Answer: A finished translation may conceal extensive testing of alternatives. - Explanation: Readers see one final wording even though the translator evaluated many alternatives.
- TACHS-R4-050 - Correct: A - Correct Answer: Multiple drafts help translators balance meaning, clarity, and tone. - Explanation: The passage explains how comparing drafts helps preserve meaning while improving natural expression and voice.

WRITTEN EXPRESSION ANSWERS

- TACHS-WE4-039 - Correct: B - Correct Answer: The photographs showed that glare and low contrast hid many numbers. - Explanation: The sentence explains what the photographs revealed and leads logically to the color comparison.
- TACHS-WE4-040 - Correct: A - Correct Answer: During a passing period, we also tested two number sizes. - Explanation: The correct revision is a complete sentence with a clear subject, verb, and modifier placement.
- TACHS-WE4-041 - Correct: A - Correct Answer: While walking through the hall, students correctly read the larger numbers more often. - Explanation: Choice A clearly identifies who read the numbers, under what condition, and with what result.
- TACHS-WE4-042 - Correct: C - Correct Answer: Sentence 7 - Explanation: The total number of lockers does not support the comparison of number color, size, and placement.
- TACHS-WE4-043 - Correct: A - Correct Answer: After a cardboard grid straightened the moving air, we suspended paper shapes from a thin rod inside the box. - Explanation: Choice A clearly connects the two setup steps without a run-on or misplaced modifier.
- TACHS-WE4-044 - Correct: C - Correct Answer: At the same fan setting, the narrow triangle swung through a smaller distance than the wide rectangle. - Explanation: Choice C states the testing condition and the measured difference precisely.
- TACHS-WE4-045 - Correct: B - Correct Answer: We plan to repeat the test with shapes cut to equal areas. - Explanation: The correct revision uses the standard verb form and clearly describes the planned test.
- TACHS-WE4-046 - Correct: B - Correct Answer: The boxes that were delivered today are in the office. - Explanation: The plural subject "boxes" correctly takes "were" and "are."
- TACHS-WE4-047 - Correct: C - Correct Answer: Walking into the lab, Maya noticed that the experiment was already running. - Explanation: Choice C clearly identifies Maya as the person walking into the lab.
- TACHS-WE4-048 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 contains the error: the

singular subject "each" requires "was."

TACHS-WE4-049 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 contains the error: the adverb "carefully" should modify "demonstrated."

TACHS-WE4-050 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 contains the error: the plural possessive form should be "editors' comments."

MATHEMATICS ANSWERS

TACHS-M4-038 - Correct: C - Correct Answer: $1/2$ - Explanation: The average of $2/5$ and $3/5$ is $5/10$, which equals $1/2$.

TACHS-M4-039 - Correct: D - Correct Answer: 72 - Explanation: Compute $2^3 \times 3^2 = 8 \times 9 = 72$.

TACHS-M4-040 - Correct: A - Correct Answer: 7 - Explanation: The next multiple of 9 after 587 is 594, and $594 - 587 = 7$.

TACHS-M4-041 - Correct: B - Correct Answer: (3, -1) - Explanation: Substitute $y = 2 - x$: $2 - x = -2x + 5$, so $x = 3$ and $y = -1$.

TACHS-M4-042 - Correct: C - Correct Answer: -8 - Explanation: Increasing x by 3 changes y by -9, so $y = 1 - 9 = -8$.

TACHS-M4-043 - Correct: A - Correct Answer: The mean changes more. - Explanation: The outlier raises the mean substantially, while the median changes only from 6 to 6.5.

TACHS-M4-044 - Correct: D - Correct Answer: $9/20$ - Explanation: There are 6 multiples of 3 and 4 multiples of 5, with 15 counted twice, so 9 of 20 numbers qualify.

TACHS-M4-045 - Correct: C - Correct Answer: 14.4 minutes - Explanation: The total time is $3 \times 12 + 2 \times 18 = 72$ minutes, and $72 \div 5 = 14.4$ minutes.

TACHS-M4-046 - Correct: B - Correct Answer: 7 centimeters - Explanation: Since $\text{area} = \text{base} \times \text{height}$, the height is $84 \div 12 = 7$ centimeters.

TACHS-M4-047 - Correct: D - Correct Answer: 35 square units - Explanation: The side lengths are 7 and 5 units, so the area is $7 \times 5 = 35$ square units.

TACHS-M4-048 - Correct: A - Correct Answer: (-4, -3) - Explanation: Reflection across the y -axis changes the sign of x while leaving y unchanged.

TACHS-M4-049 - Correct: B - Correct Answer: \$2.50 - Explanation: Divide the total cost by the weight: $\$8.75 \div 3.5 = \2.50 per pound.

TACHS-M4-050 - Correct: C - Correct Answer: 12 liters - Explanation: Multiply 0.012 by 1,000 to get 12 liters.

ABILITY ANSWERS

TACHS-A4-038 - Correct: D - Correct Answer: ■ → - Explanation: Filling □ gives ■, and rotating ↑ clockwise gives →.

TACHS-A4-039 - Correct: A - Correct Answer: ▲ - Explanation: Four triangles minus three triangles leaves one, which becomes filled: ▲.

TACHS-A4-040 - Correct: C - Correct Answer: □ - Explanation: The repeating left shift makes the final symbol a square.

TACHS-A4-041 - Correct: B - Correct Answer: Top □; Bottom ■ - Explanation: A clockwise rotation places the left open square on top and the right filled square on the bottom.

TACHS-A4-042 - Correct: D - Correct Answer: Four holes in a 2×2 arrangement - Explanation: Two perpendicular half-folds create four reflected hole positions.

TACHS-A4-043 - Correct: B - Correct Answer: Two matching notches reflected across the diagonal - Explanation: The outer-edge cut is duplicated by reflection across the diagonal fold.

TACHS-A4-044 - Correct: A - Correct Answer: Three holes in a vertical column near the left side - Explanation: All three horizontal strips are stacked, so the punch appears once in each strip at the same left position.

TACHS-A4-045 - Correct: D - Correct Answer: One circular hole centered on the horizontal midline - Explanation: The two semicircular halves reflect across the fold to form one complete circle on the midline.

TACHS-A4-046 - Correct: B - Correct Answer: ◇ ◆ ◇ - Explanation: Each valid group has matching open shapes on the outside and the filled version in the center.

TACHS-A4-047 - Correct: A - Correct Answer: ← ◇◇ - Explanation: Each arrangement has one direction arrow followed by two identical open shapes.

TACHS-A4-048 - Correct: C - Correct Answer: ◇ \ ◆ - Explanation: Each group contains an open shape, a line, and the matching filled shape in that order.

TACHS-A4-049 - Correct: C - Correct Answer: ← ↑ ↓ - Explanation: In each group, the second arrow is 90° clockwise from the first and the third is 90° counterclockwise.

TACHS-A4-050 - Correct: D - Correct Answer: Open triangle △ containing ◆ - Explanation: Each figure has an open boundary shape containing a different filled shape.